
Modern C++ for Python Programmers

From Absolute Beginner to Systems, Graphics, and Game Development

Contents

0.1	Modern C++ for Python Programmers	4
0.1.1	From Absolute Beginner to Systems, Graphics, and Game Development	4
0.1.2	How to Use This Book	4
0.1.3	Table of Contents	4
0.2	Part I – Foundations	5
0.2.1	What Does “Running a Program” Actually Mean?	5
0.2.2	How Python Runs Your Code	5
0.2.3	How C++ Runs Your Code	7
0.2.4	Your First C++ Program	9
0.2.5	Compiling: What the Flags Mean	10
0.2.6	What Is in the Compiled Binary?	11
0.2.7	Multi-File Programs	12
0.2.8	Reading Compiler Errors	12
0.2.9	Common Mistakes in This Chapter	14
0.2.10	Exercises	15
0.2.11	What IS a Variable?	16
0.2.12	How Integers Are Stored: Binary and Two’s Complement	17
0.2.13	The Fundamental Types	19
0.2.14	Fixed-Width Integer Types	21
0.2.15	Declaring and Initializing Variables	21
0.2.16	Memory Layout of Variables	23
0.2.17	auto – Type Deduction	23
0.2.18	const and constexpr	24
0.2.19	Type Conversion	25
0.2.20	Checking Sizes with sizeof	25
0.2.21	Common Mistakes in This Chapter	26
0.2.22	Exercises	27
0.2.23	Arithmetic Operators	29
0.2.24	Increment and Decrement Operators	30
0.2.25	Compound Assignment	31
0.2.26	Comparison Operators	31
0.2.27	Logical Operators	32
0.2.28	Bitwise Operators	33
0.2.29	Operator Precedence	35
0.2.30	The Ternary Operator	35
0.2.31	Common Mistakes in This Chapter	36
0.2.32	Exercises	37
0.2.33	if/else if/else	38
0.2.34	switch	39

0.2.35	while Loop	40
0.2.36	do-while Loop	41
0.2.37	for Loop: C-Style	41
0.2.38	Range-Based for Loop (C++11)	43
0.2.39	break and continue	44
0.2.40	Loop Execution Visualization	44
0.2.41	Common Mistakes in This Chapter	45
0.2.42	Exercises	47
0.2.43	What Happens When You Call a Function	49
0.2.44	Function Syntax	50
0.2.45	Declarations vs Definitions	50
0.2.46	Pass by Value: What Is Copied and Why It Matters	52
0.2.47	Default Parameters	53
0.2.48	Function Overloading	54
0.2.49	Compiler Messages for Function Errors	54
0.2.50	Common Mistakes in This Chapter	56
0.2.51	Exercises	57
0.3	Part II – Core C++ (the part that is not like Python)	59
0.3.1	The Problem References Solve	59
0.3.2	What Is a Reference?	60
0.3.3	Pass by Reference	61
0.3.4	Pass by const Reference	61
0.3.5	References Cannot Be Null	62
0.3.6	Returning References	63
0.3.7	Common Mistakes in This Chapter	63
0.3.8	Exercises	64
0.3.9	The Address of a Variable	65
0.3.10	Modifying Through a Pointer	66
0.3.11	nullptr: The Null Pointer	67
0.3.12	Pointer Arithmetic	67
0.3.13	Pointers vs Python References	68
0.3.14	The -> Operator	69
0.3.15	const and Pointers	69
0.3.16	Common Mistakes in This Chapter	70
0.3.17	Exercises	71
0.3.18	Two Regions of Memory	72
0.3.19	The Stack in Detail	72
0.3.20	The Heap in Detail	73
0.3.21	Why Python Uses the Heap for Everything	74
0.3.22	Memory Leaks	75
0.3.23	Stack vs Heap: Which to Use	76
0.3.24	Common Mistakes in This Chapter	76
0.3.25	Exercises	77
0.3.26	Why const Matters	78
0.3.27	const Variables	78
0.3.28	const Function Parameters	79
0.3.29	const Member Functions	79

0.3.30	<code>constexpr</code> – Compile-Time Constants	80
0.3.31	Cascading <code>const</code>	80
0.3.32	Common Mistakes in This Chapter	81
0.3.33	Exercises	82
0.3.34	C-Style Arrays (Understand, then Avoid)	82
0.3.35	<code>std::vector</code> – The Right Way to Do Dynamic Arrays	84
0.3.36	<code>std::string</code> – The Right Way to Handle Text	85
0.3.37	<code>std::array</code> – Fixed-Size Array With Safety	87
0.3.38	Common Mistakes in This Chapter	87
0.3.39	Exercises	88
0.3.40	Scope	90
0.3.41	Lifetime	91
0.3.42	Organizing Code: The Header/Source Split	91
0.3.43	Namespaces	93
0.3.44	Static Local Variables	95
0.3.45	Common Mistakes in This Chapter	95
0.3.46	Exercises	96

0.1 Modern C++ for Python Programmers

0.1.1 From Absolute Beginner to Systems, Graphics, and Game Development

Target Standard: C++23 | **Prerequisites:** Comfortable Python, zero C++ assumed

0.1.2 How to Use This Book

Each chapter teaches one concept from scratch. Every concept is first shown in Python (which you know), then explained in C++, then built up with multiple examples, memory diagrams where helpful, common mistakes, and exercises with answers.

Work through Part I completely before moving on. The foundations – types, memory, functions – underpin everything else.

0.1.3 Table of Contents

Part I – Foundations 1. [The Compilation Model and Your First Program](#) 2. [Variables, Types, and the Static Type System](#) 3. [Operators and Expressions](#) 4. [Control Flow: Branching and Loops](#) 5. [Functions, Overloading, and Declarations vs Definitions](#)

Part II – Core C++ (the part that is not like Python) 6. [References – Aliases for Variables](#) 7. [Pointers and Memory Addresses](#) 8. [The Stack and the Heap](#) 9. [const Correctness](#) 10. [Arrays, `std::vector`, and `std::string`](#) 11. [Scope, Lifetime, and Organizing Code into Files](#)

Part III – Ownership and Memory Management 12. [RAII – The Core Idea That Replaces Garbage Collection](#) 13. [Dynamic Allocation: `new`, `delete`, and Why You Avoid Them](#) 14. [Smart Pointers: `unique\_ptr`, `shared\_ptr`, `weak\_ptr`](#) 15. [Move Semantics, lvalues and rvalues](#) 16. [The Rule of 0, 3, and 5](#)

Part IV – Object-Oriented Programming 17. [Classes, Objects, and Encapsulation](#) 18. [Constructors, Destructors, and Initialization](#) 19. [Inheritance and Composition](#) 20. [Virtual Functions and Polymorphism](#) 21. [Abstract Classes and Interfaces](#) 22. [Operator Overloading](#)

Part V – Generic Programming 23. [Function and Class Templates](#) 24. [Template Specialization and Variadic Templates](#) 25. [Concepts \(C++20\) – Compile-Time Duck Typing](#) 26. [An Introduction to Template Metaprogramming](#)

Part VI – The Standard Library 27. [Containers: `vector`, `map`, `set`, `array`, and `friends`](#) 28. [Iterators](#) 29. [Algorithms: `sort`, `find`, `transform`, and the rest](#) 30. [Lambdas and Function Objects](#) 31. [Ranges and Views \(C++20\)](#) 32. [Utility Types: `optional`, `variant`, `any`, `tuple`](#)

Part VII – Modern C++ (C++11 to C++23) 33. [`auto`, type deduction, and structured bindings](#) 34. [`constexpr` and compile-time computation](#) 35. [`std::format` and modern string handling](#) 36. [Coroutines and Generators](#) 37. [Modules \(C++20\)](#)

Part VIII – Performance 38. [Value vs Reference Semantics](#) 39. [How Memory Layout Affects Speed](#) 40. [Data-Oriented Design](#) 41. [Profiling and Flamegraphs](#)

Part IX – Concurrency 42. [Threads and `std::jthread`](#) 43. [Mutexes, Locks, and Race Conditions](#) 44. [Atomics and the C++ Memory Model](#) 45. [Async, Futures, and Tasks](#)

Part X – Graphics and Game Development 46. [The Math: vectors, matrices, quaternions](#) 47. [How the GPU Works](#) 48. [OpenGL Fundamentals](#) 49. [Moving to Vulkan](#) 50. [Game Loop, ECS Architecture, and Engine Design](#)

Part XI – Systems Programming 51. [The Machine: registers, memory, syscalls](#) 52. [Working with the OS and Linux APIs](#) 53. [Networking from the Ground Up](#) 54. [Where C++ Meets C, eBPF, and Go](#)

Appendices - [A. Setting Up Your Toolchain](#) - [B. Compiler Flags Reference](#) - [C. Python to C++ Cheat Sheet](#) - [D. Common Mistakes and How to Debug Them](#)

0.2 Part I – Foundations

Chapter 1: The Compilation Model and Your First Program

0.2.1 What Does “Running a Program” Actually Mean?

Before you write a single line of C++, you need to understand something Python completely hides: what a computer actually does to run your code.

When you type `python hello.py`, a lot of invisible machinery kicks in. When you compile and run a C++ program, that machinery becomes visible – and you control it. That visibility is both the source of C++’s power and the main source of beginner confusion.

0.2.2 How Python Runs Your Code

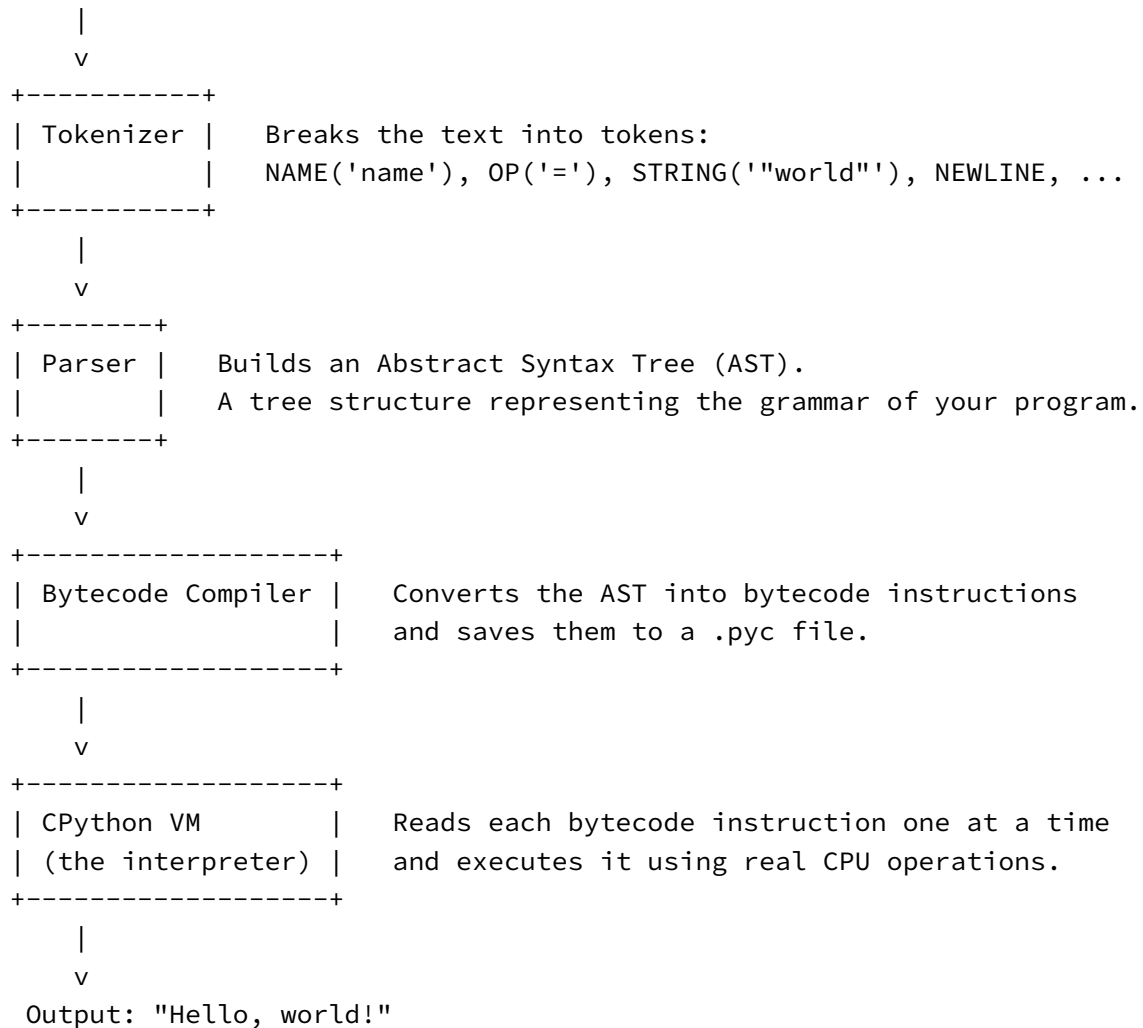
Let’s trace exactly what happens when you run a Python script.

```
# hello.py
name = "world"
print(f"Hello, {name}!")
```

```
$ python hello.py
Hello, world!
```

CPython (the standard Python interpreter) runs your code through five stages:

hello.py (plain text file)



The .pyc file contains instructions for a **virtual machine** – a software CPU that Python invented. These are NOT instructions your real CPU understands. The CPython VM translates each bytecode instruction into real CPU instructions as the program runs.

You can actually see the bytecode:

```
import dis

def greet(name):
    return f"Hello, {name}!"

dis.dis(greet)
```

Output:

```

2          0 RESUME                0

3          2 LOAD_CONST            1 ('Hello, ')
          4 LOAD_FAST              0 (name)
```

6	FORMAT_VALUE	0
8	BUILD_STRING	2
10	RETURN_VALUE	

LOAD_FAST, FORMAT_VALUE, BUILD_STRING – your CPU has never heard of these. The CPython VM reads each one and does the corresponding real work.

The consequence: There is always a middleman between your code and the CPU. Every Python operation passes through the interpreter loop. This is why Python is typically 10 to 100 times slower than C++ for CPU-bound work.

0.2.3 How C++ Runs Your Code

C++ uses a completely different model. Your source code is **translated directly into CPU instructions** before the program runs. There is no interpreter, no virtual machine, no middleman. When you run the program, your CPU executes your instructions directly.

This is called **ahead-of-time compilation**.

The translation is done by a chain of three programs:

```
hello.cpp (plain text, C++ source)
  |
  v
+-----+
| Preprocessor | Handles lines starting with #.
|              | Pastes #include files, substitutes #define macros.
|              | Pure text manipulation -- does NOT understand C++.
+-----+
  |
  | (preprocessed .cpp -- still text, but with all #includes pasted in)
  v
+-----+
| Compiler     | Parses C++, checks all types, optimizes, generates
|              | machine code for your CPU architecture.
+-----+
  |
  | (object file: .o -- real machine code, but references unresolved)
  v
+-----+
| Linker       | Combines all .o files plus library code into one
|              | self-contained executable.
+-----+
  |
  v
./hello (executable -- native machine code, runs directly on CPU)
```


0.2.3.1 Stage 1: The Preprocessor in Detail

The preprocessor is a dumb text tool. It knows only three things:

- `#include <file>` or `#include "file"` – paste the entire text of that file here
- `#define NAME text` – replace every occurrence of `NAME` with `text`
- `#ifdef` / `#ifndef` / `#else` / `#endif` – include or exclude blocks of text

It does not parse C++. It does not understand types. It just manipulates text.

When you write `#include <iostream>`, the preprocessor pastes thousands of lines into your file before the compiler sees anything. You can see how much:

```
$ g++ -E hello.cpp | wc -l
45231
```

One line expands to 45,231 lines. That is why large projects are slow to compile – every `.cpp` file re-expands every header it includes.

0.2.3.2 Stage 2: The Compiler in Detail

The compiler reads the preprocessed text and does the real work:

1. **Parsing:** builds an AST (same concept as Python's parser)
2. **Semantic analysis:** checks types, resolves names, catches errors
3. **Optimization** (with `-O2`): rewrites your code to run faster
4. **Code generation:** emits machine code for your target CPU

The output is an **object file** (`.o`). It contains real machine code, but it has gaps – references to functions defined in other files (like `std::cout`) that haven't been connected yet.

0.2.3.3 Stage 3: The Linker in Detail

The linker takes all object files and the standard library and combines them into one executable. It resolves every function call to an actual memory address. When `main.cpp` calls `std::cout`, the linker finds `cout`'s implementation in the standard library and wires them together.

0.2.3.4 The Core Difference

Python:

source --> bytecode (at first run) --> CPU (via interpreter, every run)

C++:

source --> machine code (once, at compile time) --> CPU (directly, every run)

After you compile once, running the program has zero translation overhead. The CPU just executes instructions.

0.2.4 Your First C++ Program

Create a file called `hello.cpp`:

```
#include <iostream>

int main() {
    std::cout << "Hello, world!\n";
    return 0;
}
```

Compile and run:

```
$ g++ -std=c++23 -o hello hello.cpp
$ ./hello
Hello, world!
```

Now let's understand every token in this program.

0.2.4.1 `#include <iostream>`

A preprocessor directive. Pastes the `iostream` header (thousands of lines) into your file.

`iostream` declares `std::cout`, the standard output stream. Without it:

```
error: 'cout' is not a member of 'std'
  4 |     std::cout << "Hello, world!\n";
    |     ^~~~~~
note: 'std::cout' is defined in header '<iostream>';
      did you forget to '#include <iostream>'?
```

Angle brackets `<iostream>` mean “find this in the compiler’s system include path.” Double quotes `"my-file.h"` mean “look in my project directory first.”

0.2.4.2 `int main()`

The **entry point** of every C++ program. When the OS launches your executable, it calls `main()`. Everything begins here.

`int` is the **return type**. The function returns an integer to the OS as an exit code: `-0` = success (by convention) - Non-zero = some kind of failure

Unlike Python where top-level code runs directly, C++ requires exactly one function called `main` as the entry point. You can’t rename it or have two of them.

The `{}` braces delimit the function body, playing the same role as `:` and indentation in Python.

0.2.4.3 `std::cout << "Hello, world!\n";`

This is how you print to the terminal.

std is a **namespace** – a named container for related identifiers. The entire C++ standard library lives inside `std`. The `::` operator means “look inside.” So `std::cout` means “the `cout` that lives inside the `std` namespace.”

cout stands for “character output.” It is an object representing your terminal’s standard output.

`<<` is the **stream insertion operator**. It sends the value on the right into the stream on the left. You can chain it:

```
std::cout << "Hello" << ", " << "world" << "!\n";
```

Each `<<` sends one more thing into the stream.

"Hello, world!\n" is a string literal. The `\n` is a newline character. Unlike Python’s `print()`, which adds a newline automatically, C++ streams do not add newlines unless you include them.

; terminates the statement. Every statement ends with a semicolon. This is the single most common beginner error. Forgetting it produces an error on the *next* line because the compiler doesn’t notice until it sees an unexpected token.

0.2.4.4 `return 0;`

Returns the exit code 0 (success) to the OS. In `main` specifically, you can omit this and the compiler inserts it, but writing it is explicit and clear.

0.2.5 Compiling: What the Flags Mean

```
$ g++ -std=c++23 -Wall -Wextra -o hello hello.cpp
```

Flag	What it does
<code>g++</code>	The GNU C++ compiler. Also try <code>clang++</code> (often better error messages).
<code>-std=c++23</code>	Use the C++23 standard. Without this you get an older, less capable standard.
<code>-Wall</code>	Enable all common warnings. The W stands for Warning.
<code>-Wextra</code>	Enable even more warnings.
<code>-o hello</code>	Name the output executable <code>hello</code> . Default without <code>-o</code> is <code>a.out</code> .

Flag	What it does
hello.cpp	The source file to compile.

0.2.5.1 Two Build Modes You Will Use

```
# Development (use this while writing code):
$ g++ -std=c++23 -Wall -Wextra -g -fsanitize=address,undefined -o hello hello.cpp
#                                     |   |
#                                     |   +-- catches memory bugs and UB at runtime
#                                     +----- adds debug symbols for stack traces

# Release (use this for final builds):
$ g++ -std=c++23 -O2 -o hello hello.cpp
#                                     |
#                                     +-- enables optimization (2x-10x faster)
```

The sanitizers (`-fsanitize=address,undefined`) catch entire classes of bugs that would otherwise produce silent wrong answers or random crashes. Always use them during development.

0.2.6 What Is in the Compiled Binary?

The executable file is organized into sections:

`./hello` (ELF executable on Linux, Mach-O on macOS, PE on Windows)

Section	Contents
-----	-----
<code>.text</code>	Your compiled machine code instructions
<code>.rodata</code>	Read-only data: string literals like "Hello, world!\n"
<code>.data</code>	Initialized global variables
<code>.bss</code>	Uninitialized global variables (zeroed at program start)

You can disassemble the executable to see what your code became:

```
$ objdump -d -C hello | grep -A 15 "<main>"
```

```
0000000000001149 <main>:
   1149: push    rbp
   114a: mov     rbp, rsp
   114d: lea     rdi, [rip+0xeb0]    <- loads address of "Hello, world!\n"
   1154: call    1060 <puts@plt>    <- calls puts() (compiler optimized cout to this)
   1159: mov     eax, 0x0
   115e: pop     rbp
   115f: ret
```

Seven CPU instructions. No interpreter, no type checking, no reference counting. This is what runs when you execute `./hello`.

0.2.7 Multi-File Programs

Once programs grow beyond one file, you split them:

```
main.cpp  --compile-->  main.o  --+
                                     +--> linker --> program
greet.cpp --compile--> greet.o  --+
```

```
// greet.h -- declaration (the interface, what other files need to know)
#pragma once
#include <string>
void greet(const std::string& name);

// greet.cpp -- definition (the actual implementation)
#include <iostream>
#include "greet.h"
void greet(const std::string& name) {
    std::cout << "Hello, " << name << "!\n";
}

// main.cpp -- uses the function
#include "greet.h"
int main() {
    greet("world");
    return 0;
}
```

```
$ g++ -std=c++23 -o program main.cpp greet.cpp
```

Chapter 11 covers the full rationale for this split.

0.2.8 Reading Compiler Errors

The compiler is your most important tool. It catches errors before your program runs. Learning to read its output quickly is a core skill.

0.2.8.1 Missing Semicolon

```
int main() {
    std::cout << "Hello"    // missing semicolon here
    return 0;
}
```

```
hello.cpp:4:5: error: expected ';' before 'return'
   4 |     std::cout << "Hello"
     |                                     ^
     |                                     ;
   5 |     return 0;
```

The caret ^ marks where the problem was detected – the end of line 4. The message says “before ‘return’” because the compiler only noticed something was wrong when it hit `return`.

0.2.8.2 Missing #include

```
int main() {
    std::cout << "Hello\n";
}
```

```
hello.cpp:2:5: error: 'cout' is not a member of 'std'
   2 |     std::cout << "Hello\n";
     |     ^~~~~~
note: 'std::cout' is defined in header '<iostream>';
     did you forget to '#include <iostream>'?
```

Read every note: line. Modern compilers often tell you exactly what to do.

0.2.8.3 Undefined Reference (Linker Error)

```
// main.cpp
void greet(const std::string& name); // declared but never defined anywhere

int main() { greet("world"); }
```

```
/usr/bin/ld: main.o: undefined reference to `greet(std::string const&)'
collect2: error: ld returned 1 exit status
```

“Undefined reference” means the compiler accepted the code (it saw the declaration), but the linker couldn’t find the function’s body. Either you forgot to define it, or forgot to include its `.cpp` file in the build command.

0.2.8.4 Type Mismatch

```
int x = "hello";
```

error: invalid conversion from 'const char*' to 'int'

```
1 | int x = "hello";
  |           ^~~~~~
```

The static type system caught this before the program ran. Python would allow `x = 5; x = "hello"` – C++ refuses.

0.2.9 Common Mistakes in This Chapter

0.2.9.1 Mistake 1: Forgetting the semicolon

The bug: `std::cout << "Hello"` with no `;`

The symptom: Error message appears on the *next* line, not the line where you forgot it. The compiler doesn't notice until it encounters an unexpected token.

The fix: Every statement ends with `;`. After `return 0`. After every `cout`. After every variable declaration.

0.2.9.2 Mistake 2: Writing `cout` without `std::`

The bug:

```
cout << "Hello\n"; // 'cout' -- where is it?
```

The symptom:

error: 'cout' was not declared in this scope

The fix: Write `std::cout`. You can also add `using std::cout;` at the top of a function to avoid typing `std::` everytime. Do not use `using namespace std;` – it pulls in hundreds of names that may conflict with your own.

0.2.9.3 Mistake 3: Confusing `#include <...>` with `#include "..."`

The bug: Writing `#include "iostream"` instead of `#include <iostream>`

What happens: The preprocessor looks for `iostream` in the current directory first. It may find nothing and fail, or find the wrong file.

The fix: Use `<...>` for system and library headers. Use `"..."` for your own headers.

0.2.10 Exercises

Exercise 1.1 – Predict the output

What does this print? Work it out before running it.

```
#include <iostream>
int main() {
    std::cout << "Line 1\n";
    std::cout << "Line " << 2 << "\n";
    std::cout << "Line " << 1 + 2 << "\n";
    return 0;
}
```

Answer:

Line 1
Line 2
Line 3

1 + 2 is evaluated to 3 before being inserted into the stream.

Exercise 1.2 – Find all three bugs

This program has exactly three syntax errors. Find them all.

```
#include <iostream>
int main() {
    std::cout << "Hello, world!"
    std::cout << "How are you?"
    return 0
}
```

Answer: 1. Missing ; after "Hello, world!" 2. Missing ; after "How are you?" 3. Missing ; after return 0

Exercise 1.3 – Write from scratch

Write a program that prints this output exactly (copy the spacing):

```
=== My C++ Program ===
Author: Your Name
Version: 1.0
```

Answer:

```
#include <iostream>
int main() {
    std::cout << "=== My C++ Program ===\n";
    std::cout << "Author: Your Name\n";
    std::cout << "Version: 1.0\n";
    return 0;
}
```

Exercise 1.4 – Explain the difference

In your own words: what is the difference between compiling with `-O0` and `-O2`? Why would you use each?

Answer: `-O0` (no optimization) translates your code nearly literally into machine code. The machine code closely mirrors your source, making it easier to debug – variables are in the places you’d expect, and the program steps through code the way you wrote it. Use this during development.

`-O2` enables aggressive optimization: the compiler may inline functions, eliminate redundant computations, reorder instructions, and more. The program runs significantly faster. The machine code may look very different from your source. Use this for release builds.

Chapter 2: Variables, Types, and the Static Type System

0.2.11 What IS a Variable?

In Python, a variable is a **name tag** attached to an object. The object lives on the heap; the name is just a reference to it.

```
x = 42          # 'x' is a label pointing to an int object on the heap
y = x          # 'y' is another label pointing to the SAME object
x = "hello"    # 'x' now points to a different object; y is unchanged
print(y)       # 42
```

The object carries its own type:

Python memory:

Variable	Points to
-----	-----
x ----->	[type: int refcount: 2 value: 42]
y ----->	[same object]

After `x = "hello"`:

x ----->	[type: str refcount: 1 value: "hello"]
y ----->	[type: int refcount: 1 value: 42]

Every Python variable is a pointer. The object it points to carries its type, a reference count, and metadata.

In C++, a variable is a **named region of memory with a fixed type**. The variable IS the storage. There is no separate heap object (unless you explicitly create one), no reference count, no type tag at runtime.

```
int x = 42; // 'x' is 4 bytes of memory containing the bit pattern for 42
           // The type 'int' exists only in the compiler's mind
```

C++ memory:

Address	Variable	Bytes stored
-----	-----	-----
0x7fff10	x (int)	[00][00][00][2A] <- that is 42 in hex

No type tag. No reference count. No metadata. Four bytes. That is it.

The type `int` tells the **compiler** (not the CPU, not the runtime): - Reserve 4 bytes - Treat those bytes as a signed two's complement integer - Allow operations like `+`, `-`, `*`, `/` - Reject operations like string concatenation

At runtime, there are only bytes. Types are a compile-time concept.

This is the most fundamental difference between Python and C++. Almost every other difference you will encounter flows from this one.

0.2.12 How Integers Are Stored: Binary and Two's Complement

To understand C++ types, you need to understand how integers are stored in memory. If you already know two's complement, skim this section.

A computer stores everything as bits: 0 or 1. Eight bits make one byte.

The number 42 in binary:

$$42 = 32 + 8 + 2 = 2^5 + 2^3 + 2^1$$

Bit positions (0 = rightmost):

7	6	5	4	3	2	1	0
0	0	1	0	1	0	1	0
		32		8		2	
0							

$$0 \times 128 + 0 \times 64 + 1 \times 32 + 0 \times 16 + 1 \times 8 + 0 \times 4 + 1 \times 2 + 0 \times 1 = 42$$

An `int` uses 4 bytes = 32 bits. So 42 stored as a 32-bit int:

Byte 3	Byte 2	Byte 1	Byte 0
00000000	00000000	00000000	00101010
= 42			

No type metadata anywhere. The CPU interprets these bytes according to which instruction operates on them.

0.2.12.1 How Negative Numbers Work: Two's Complement

For signed integers, the highest bit is the sign bit: 0 = positive, 1 = negative.

For a 32-bit signed int: - Maximum positive: 0111 1111 ... 1111 = 2,147,483,647 - Maximum negative: 1000 0000 ... 0000 = -2,147,483,648 - -1: 1111 1111 ... 1111 (all bits set)

To negate a number: flip all bits, then add 1.

42 in binary:	0000 0000 0000 0000 0000 0000 0010 1010
Flip all bits:	1111 1111 1111 1111 1111 1111 1101 0101
Add 1:	1111 1111 1111 1111 1111 1111 1101 0110 = -42

The critical implication: **signed integer overflow is undefined behavior in C++.**

```
int max = 2147483647;    // the largest possible int
int bad = max + 1;       // UNDEFINED BEHAVIOR
                        // The standard does not define what happens.
                        // In practice: wraps to -2147483648 on most systems.
                        // But the compiler is allowed to assume overflow
                        // never happens and optimize in ways that break you.
```

Python integers are arbitrary precision – they never overflow. C++ `int` has fixed size and hard limits. Always think about whether your values can exceed the range.

0.2.13 The Fundamental Types

0.2.13.1 bool

```
bool yes = true;
bool no  = false;

std::cout << yes << "\n"; // prints 1
std::cout << no  << "\n"; // prints 0
```

Stores true/false. Takes 1 byte (even though 1 bit would suffice – byte is the smallest addressable unit on most hardware). `true` is stored as 1, `false` as 0.

Any non-zero integer converts to `true`, zero converts to `false`:

```
bool b1 = 42; // true  (42 != 0)
bool b2 = 0;  // false
bool b3 = -1; // true  (-1 != 0)
```

0.2.13.2 char

```
char letter = 'A'; // single character, single quotes
char newline = '\n'; // escape sequences work like Python
char tab = '\t';
char null_c = '\0'; // null character (value 0)
```

One byte. Holds one ASCII character. The character is stored as its ASCII code: `'A'` = 65, `'Z'` = 90, `'0'` = 48, `'\n'` = 10.

You can do arithmetic on chars because they are just numbers:

```
char c = 'A';
std::cout << c + 1; // 66 (int arithmetic)
std::cout << static_cast<char>(c + 1); // 'B' (cast back to char)
std::cout << (char)('a' + 2); // 'c' (C-style cast, avoid)
```

0.2.13.3 Integer Types

Type	Typical size	Signed range	Notes
<code>short</code>	2 bytes	-32,768 to 32,767	Rarely used
<code>int</code>	4 bytes	-2.1B to 2.1B	Default integer type
<code>long</code>	4 or 8 bytes	platform-dependent	Avoid – use long long
<code>long long</code>	8 bytes	-9.2e18 to 9.2e18	Use when int overflows

To make a type unsigned (no negatives, larger maximum):

```
unsigned int u = 4000000000u; // 'u' suffix = unsigned literal
unsigned long long big = 18446744073709551615ull; // 'ull' suffix
```

An unsigned type with N bits holds 0 to $2^N - 1$. A signed type with N bits holds $-2^{(N-1)}$ to $2^{(N-1)} - 1$.

0.2.13.4 float and double

```
float f = 3.14f; // 32-bit floating point, 'f' suffix is required
double d = 3.14; // 64-bit floating point -- use this by default
```

double is what Python calls float. Use double unless you have a specific reason (memory, GPU/SIMD code).

Why prefer double? Precision: - float: approximately 7 significant decimal digits - double: approximately 15 significant decimal digits

```
float f = 1.23456789f;
double d = 1.23456789;
std::cout << std::setprecision(10) << f << "\n"; // 1.234567881 (wrong at 7th digit)
std::cout << std::setprecision(10) << d << "\n"; // 1.23456789 (correct)
```

Floating point is not exact – same in Python and C++:

```
# Python
print(0.1 + 0.2) # 0.30000000000000004
```

```
// C++
#include <iomanip>
double x = 0.1 + 0.2;
std::cout << x << "\n"; // 0.3 (default rounds nicely)
std::cout << std::setprecision(17) << x << "\n"; // 0.30000000000000004
std::cout << std::setprecision(17) << 0.3 << "\n"; // 0.29999999999999999
```

0.1, 0.2, and 0.3 cannot be represented exactly in binary floating point – the same way $1/3$ cannot be represented exactly in decimal. This is not a C++ bug; it is IEEE 754 floating point. The implication: **never compare floating-point numbers with ==.**

0.2.13.5 std::string

```
#include <string>
std::string name = "Alice";
std::string greeting = "Hello, " + name + "!"; // concatenation with +
std::cout << greeting.size() << "\n"; // 13
```

Not a primitive type. `std::string` is a class from the standard library that manages heap-allocated character data. We cover it in depth in Chapter 10.

0.2.14 Fixed-Width Integer Types

The sizes of `short`, `int`, `long` depend on the platform and compiler. For code that must work correctly regardless of platform – networking, file formats, hardware interfaces – use exact-size types from `<cstdint>`:

```
#include <cstdint>

int8_t  a = -128;           // exactly 8 bits, signed (-128 to 127)
uint8_t b = 255;           // exactly 8 bits, unsigned (0 to 255)
int16_t c = 32767;         // exactly 16 bits, signed
uint16_t d = 65535;        // exactly 16 bits, unsigned
int32_t e = -1;            // exactly 32 bits, signed
uint32_t f = 4294967295u;   // exactly 32 bits, unsigned
int64_t g = -1LL;          // exactly 64 bits, signed
uint64_t h = 0xFFFFFFFFFFFFFFFFull; // exactly 64 bits, unsigned, all bits set
```

Use these whenever the exact bit width matters. Use `int` when it does not.

0.2.15 Declaring and Initializing Variables

0.2.15.1 Never Leave a Variable Uninitialized

This is the most important rule in this chapter:

```
int sum; // BAD: sum contains whatever bytes were
for (int i = 0; i < 10; ++i) // in that memory location before.
    sum += i; // Adding to garbage = garbage.
std::cout << sum << "\n"; // Prints something unpredictable.

int sum{0}; // GOOD: explicitly initialized to zero.
for (int i = 0; i < 10; ++i)
    sum += i;
std::cout << sum << "\n"; // 45, always.
```

On some runs, the garbage value might happen to be 0, making the bug invisible. On other runs, it will be -858993460 or similar. Undefined behavior is unpredictable.

0.2.15.2 The Four Initialization Forms

C++ has four ways to initialize a variable. They have subtle differences.

Form 1: Default initialization (leaves value unset – dangerous)

```
int x;           // x has garbage value
double d;        // d has garbage value
```

Do not do this for primitive types.

Form 2: Copy initialization (C-style, from C)

```
int x = 5;
double d = 3.14;
std::string s = "hello";
```

Looks like assignment but is initialization (happens at construction, not after). Fine for simple cases.

Form 3: Direct initialization

```
int x(5);
double d(3.14);
std::string s("hello");
```

More verbose for primitives but is the natural form for class objects with constructors.

Form 4: Brace (uniform) initialization – prefer this

```
int x{5};
double d{3.14};
std::string s{"hello"};
```

Why prefer brace initialization? It prevents **narrowing conversions** – type truncations that silently discard data:

```
// With = initialization, narrowing is SILENT:
int a = 3.7;           // truncates to 3. No error. Potential bug hiding.
int b = 300000;        // fits in int, fine
int c = 300000LL;      // long long to int, may truncate on 16-bit systems, no warning

// With {} initialization, narrowing is a COMPILE ERROR:
int a{3.7};           // error: narrowing conversion of '3.7e+0' from 'double' to 'int'
int b{300000};        // fine
int c{300000LL};      // fine (value fits in int)
```

The compiler tells you about the truncation at compile time instead of letting it silently corrupt your data at runtime.

You can value-initialize (set to zero/false/null) with empty braces:

```
int    n{};    // 0
double d{};    // 0.0
bool   b{};    // false
char   c{};    // '\0'
```

0.2.16 Memory Layout of Variables

```
bool   a{true};    // 1 byte
char   b{'X'};     // 1 byte
int    c{42};      // 4 bytes
double d{3.14};    // 8 bytes
```

Stack memory (addresses are illustrative):

Addr	Var	Size	Raw bytes (hex)	
0x1000	a (bool)	1 byte	[01]	
0x1001	b (char)	1 byte	[58]	'X' = ASCII 88 = 0x58
0x1004	c (int)	4 bytes	[00][00][00][2A]	42 = 0x2A
0x1008	d (double)	8 bytes	[40][09][1E][B8][51][EB][85][1F]	

(The actual layout may differ due to alignment, but the sizes are as shown.)

There is no type information stored at runtime. The CPU knows to treat `c` as a signed integer because the compiled instructions operating on it are integer instructions. The type system is entirely a compile-time construct.

0.2.17 auto – Type Deduction

When the type is obvious from the right-hand side, `auto` lets the compiler deduce it:

```
auto x = 42;           // int    (integer literals default to int)
auto y = 42L;          // long   (L suffix)
auto z = 42LL;         // long long (LL suffix)
auto d = 3.14;         // double (decimal literals default to double)
auto f = 3.14f;        // float  (f suffix)
auto b = true;         // bool
auto s = std::string{"hi"}; // std::string
```

`auto` does NOT make C++ dynamically typed. The type is deduced once at compile time and then fixed permanently:


```
auto x = 42;           // x is int, fixed forever
x = 3.14;             // fine: double 3.14 truncates to int 3 (with = init)
x = "hello";          // COMPILE ERROR: cannot assign const char* to int
```

auto strips top-level const and references from the deduced type. Add them explicitly if you need them:

```
int n = 5;
auto a = n;           // int    (copy)
auto& b = n;           // int&   (reference to n)
const auto& c = n;     // const int& (const reference, no copy, no modify)
```

Where auto really shines is avoiding verbose type names:

```
// Without auto:
std::vector<std::pair<std::string, int>>::iterator it = scores.begin();

// With auto (same type, more readable):
auto it = scores.begin();
```

0.2.18 const and constexpr

0.2.18.1 const – Constant After Initialization

```
const int MAX = 8;
MAX = 10;           // COMPILE ERROR: assignment of read-only variable

const int n = get_input(); // value known only at runtime -- still valid const
```

Use const for values that are initialized once and never change. The compiler enforces this – it's not just a convention.

0.2.18.2 constexpr – Must Be Computed at Compile Time

```
constexpr int BOARD = 8;
constexpr double PI = 3.14159265358979;

int grid[BOARD][BOARD]; // OK -- array size must be a compile-time constant

constexpr int square(int x) { return x * x; }
constexpr int area = square(BOARD); // 64, computed at compile time, zero runtime cost
```

`constexpr` is stronger than `const`. The value must be computable without running the program. If it isn't, you get a compile error – which is what you want.

Rule: use `constexpr` for mathematical constants, array sizes, and configuration values. Use `const` for values set at runtime that shouldn't change afterward.

0.2.19 Type Conversion

0.2.19.1 Safe Widening (Automatic)

```
int    i = 42;
double d = i;    // int to double: no data loss, happens automatically
long long l = i; // int to long long: always safe
```

0.2.19.2 Narrowing (Dangerous Without Braces)

```
double d = 3.99;
int    i = d;    // silently truncates to 3 (with = initialization)
int    j{d};     // COMPILE ERROR: narrowing (with {} initialization)
```

0.2.19.3 Explicit Conversion with `static_cast`

When you intend to convert, be explicit about it:

```
double d = 3.99;
int i = static_cast<int>(d);    // 3, explicit truncation, intent is documented

int a = 5, b = 2;
// Bug: division happens as int first, then converts to double
double wrong = a / b;           // 2.0

// Fix: cast one operand to double before dividing
double correct = static_cast<double>(a) / b; // 2.5
```

`static_cast<NewType>(expr)` is the C++ way to convert. It is checked at compile time. Do not use C-style casts like `(int)x` – they bypass some safety checks and are harder to search for in code.

0.2.20 Checking Sizes with `sizeof`

```
#include <iostream>
int main() {
    std::cout << "bool:      " << sizeof(bool)      << " byte(s)\n"; // 1
    std::cout << "char:      " << sizeof(char)      << " byte(s)\n"; // 1
    std::cout << "int:       " << sizeof(int)       << " byte(s)\n"; // 4
    std::cout << "long long: " << sizeof(long long) << " byte(s)\n"; // 8
    std::cout << "float:     " << sizeof(float)     << " byte(s)\n"; // 4
    std::cout << "double:    " << sizeof(double)    << " byte(s)\n"; // 8
    return 0;
}
```

sizeof is evaluated at compile time – zero runtime overhead. It works on both types (sizeof(int)) and variables (sizeof(x)).

0.2.21 Common Mistakes in This Chapter

0.2.21.1 Mistake 1: Uninitialized Variable

The bug:

```
int total;
for (int i = 1; i <= 5; ++i)
    total += i; // total starts with garbage, not 0
std::cout << total;
```

The symptom: Wrong or unpredictable output. Sometimes appears correct (when garbage happens to be 0).

The fix: `int total{0};`

How to detect: Compile with `-Wall` (warns about “total may be used uninitialized”). Run with `-fsanitize=undefined` for runtime detection.

0.2.21.2 Mistake 2: Signed Integer Overflow

The bug:

```
int seconds_per_year = 365 * 24 * 60 * 60; // 31,536,000 -- fits in int
int seconds_per_century = seconds_per_year * 100; // 3,153,600,000 -- OVERFLOW!
// int max is ~2.1 billion; 3.15 billion overflows
```

The symptom: Wrong answer (often a negative number). No compile error.

The fix: `long long seconds_per_century = (long long)seconds_per_year * 100;`

How to detect: `-fsanitize=undefined` reports “signed integer overflow.”

0.2.21.3 Mistake 3: Integer Division When Expecting Decimal

The bug:

```
double average = (100 + 75 + 90) / 3; // 88 -- not 88.333...
```

All literals are `int`. The division is integer division before the result is stored in `double`.

The fix: `double average = (100 + 75 + 90) / 3.0;` or use `static_cast<double>`.

0.2.21.4 Mistake 4: Comparing Floats with `==`

The bug:

```
double x = 0.1 + 0.2;
if (x == 0.3) {
    std::cout << "equal\n"; // never prints
}
```

The fix:

```
#include <cmath>
if (std::abs(x - 0.3) < 1e-9) {
    std::cout << "equal\n"; // works
}
```

0.2.22 Exercises

Exercise 2.1 – Type sizes

Without looking anything up, predict the output of this program on a 64-bit Linux system, then verify:

```
#include <iostream>
int main() {
    std::cout << sizeof(bool) << "\n";
    std::cout << sizeof(char) << "\n";
    std::cout << sizeof(short) << "\n";
    std::cout << sizeof(int) << "\n";
    std::cout << sizeof(long) << "\n";
    std::cout << sizeof(long long) << "\n";
    std::cout << sizeof(float) << "\n";
    std::cout << sizeof(double) << "\n";
}
```

Answer: 1 1 2 4 8 8 4 8 on 64-bit Linux. Note that `long` is 8 bytes on 64-bit Linux but 4 bytes on 64-bit Windows – another reason to use `long long` when you need 64 bits.

Exercise 2.2 – Brace initialization rejection

Which of these lines would brace initialization reject with a compile error?

```
int a{3.0}; // (a)
int b{3}; // (b)
int c{3LL}; // (c) LL = long long literal
double d{3}; // (d)
float e{3.14}; // (e)
```

Answer: - (a): double -> int narrows (loses the decimal). **ERROR.** - (b): int -> int, exact. Fine. - (c): long long -> int may narrow if value doesn't fit. **ERROR** (even though 3 fits, the types differ). - (d): int -> double widens (no data loss). Fine. - (e): double -> float narrows (loses precision). **ERROR.**

Exercise 2.3 – Fix the division

Fix this code so it prints 2.5 instead of 2:

```
#include <iostream>
int main() {
    int a = 5, b = 2;
    std::cout << a / b << "\n";
}
```

Answer:

```
std::cout << static_cast<double>(a) / b << "\n";
// or:
std::cout << a / static_cast<double>(b) << "\n";
// or:
std::cout << a / 2.0 << "\n";
```

Exercise 2.4 – Max value calculation

What is the maximum value of `uint32_t`? Show the calculation. Do the same for `int32_t`.

Answer: - `uint32_t`: 32 bits, unsigned. Range = 0 to $2^{32} - 1 = \mathbf{4,294,967,295}$. - `int32_t`: 32 bits, signed. Range = -2^{31} to $2^{31} - 1 = \mathbf{-2,147,483,648}$ to $\mathbf{2,147,483,647}$. One bit is used for the sign.

Chapter 3: Operators and Expressions

0.2.23 Arithmetic Operators

Most arithmetic operators work the same as Python. Division is the exception that trips everyone up.

```
int a = 17, b = 5;

std::cout << a + b << "\n"; // 22 -- addition
std::cout << a - b << "\n"; // 12 -- subtraction
std::cout << a * b << "\n"; // 85 -- multiplication
std::cout << a / b << "\n"; // 3 -- WATCH OUT: integer division
std::cout << a % b << "\n"; // 2 -- remainder (modulo)
```

The division rule: **when both operands are integers, / discards the remainder.**

```
# Python: / always returns a float
print(17 / 5) # 3.4
print(17 // 5) # 3 (floor division, rounds toward negative infinity)
print(-17 // 5) # -4 (floor: -3.4 rounds down to -4)
```

```
// C++: / on integers truncates toward zero (not floor)
std::cout << 17 / 5 << "\n"; // 3 (truncated toward zero)
std::cout << -17 / 5 << "\n"; // -3 (truncated toward zero, NOT -4)
std::cout << 17.0 / 5 << "\n"; // 3.4 (float division: 17.0 is a double)
```

The key distinction: C++ truncates toward zero. Python's `//` floors toward negative infinity. For positive numbers they agree. For negative numbers they differ.

0.2.23.1 The Classic Division Trap

```
int numerator = 7;
int denominator = 2;

double result = numerator / denominator; // 3.0, NOT 3.5
```

What happened? The division `numerator / denominator` is evaluated first – both are `int`, so integer division gives 3. Then 3 is converted to 3.0 for the `double` assignment.

The fix: make at least one operand a `double` before the division:

```
double result = static_cast<double>(numerator) / denominator; // 3.5
// or:
double result = numerator / static_cast<double>(denominator); // 3.5
// or (if one is a literal):
double result = numerator / 2.0; // 3.5
```

0.2.23.2 Modulo With Negative Numbers

```
std::cout << 7 % 3 << "\n"; // 1 (7 = 2*3 + 1)
std::cout << -7 % 3 << "\n"; // -1 (sign follows dividend in C++)
std::cout << 7 % -3 << "\n"; // 1 (sign follows dividend)
```

Python's `%` always returns non-negative when the divisor is positive (e.g., `-7 % 3 = 2` in Python). C++'s `%` sign follows the dividend.

This matters for “wrap around” patterns. To get Python-like non-negative modulo:

```
int python_mod(int a, int b) {
    return ((a % b) + b) % b;
}
// python_mod(-7, 3) = ((-1) + 3) % 3 = 2 % 3 = 2 -- same as Python
```

0.2.24 Increment and Decrement Operators

C++ has `++` (add 1) and `--` (subtract 1) as standalone operators. Python does not.

```
int x = 5;
++x; // pre-increment: x becomes 6
x++; // post-increment: x becomes 7
--x; // pre-decrement: x becomes 6
x--; // post-decrement: x becomes 5
```

Used as standalone statements, `++x` and `x++` are identical in effect. The difference appears when used inside a larger expression:

```
int x = 5;
int a = ++x; // pre: x becomes 6 FIRST, then a = 6 (new value)
int b = x++; // post: b = 6 (old value), then x becomes 7
```

Pre-increment (++x):

Step 1: increment x (x = 7)

Step 2: expression evaluates to new x (7)

Post-increment (x++):

Step 1: save current x (6)

Step 2: increment x (x = 7)

Step 3: expression evaluates to saved old x (6)

Always prefer ++x (pre-increment) as a habit. It is never slower than x++, and for iterator objects (Chapter 28) it can be significantly faster because post-increment must save a copy of the old value.

0.2.25 Compound Assignment

These modify a variable in place:

```
int x = 20;
x += 5;    // x = 25    (same as x = x + 5)
x -= 3;    // x = 22
x *= 2;    // x = 44
x /= 4;    // x = 11    (integer division if x is int)
x %= 3;    // x = 2      (remainder)
x <<= 2;   // x = 8      (left shift by 2: same as x * 4)
x >>= 1;   // x = 4      (right shift by 1: same as x / 2 for non-negative)
```

0.2.26 Comparison Operators

All return bool:

```
int a = 5, b = 10;
bool r1 = (a == b);    // false -- equal
bool r2 = (a != b);    // true  -- not equal
bool r3 = (a < b);     // true  -- less than
bool r4 = (a > b);     // false -- greater than
bool r5 = (a <= b);    // true  -- less than or equal
bool r6 = (a >= b);    // false -- greater than or equal
```


0.2.26.1 The Floating-Point Comparison Problem

```
double x = 0.1 + 0.2;
double y = 0.3;
std::cout << (x == y) << "\n";    // 0 (false!)
```

0.1 cannot be represented exactly in binary. Neither can 0.2 or 0.3. The sum 0.1 + 0.2 comes out as 0.30000000000000004... Stored 0.3 is 0.29999999999999998... They are not equal, even though mathematically they should be.

This is not a C++ problem – the same thing happens in Python and every other language that uses IEEE 754 floating point (which is all of them).

Rule: Never compare floating-point numbers with ==. Use a tolerance.

```
#include <cmath>    // for std::abs

// Returns true if a and b are within tolerance of each other
bool nearly_equal(double a, double b, double tolerance = 1e-9) {
    return std::abs(a - b) < tolerance;
}

if (nearly_equal(0.1 + 0.2, 0.3)) {
    std::cout << "equal (within tolerance)\n"; // this prints
}
```

Choose tolerance based on the scale of your values: - For values around 1.0: 1e-9 is reasonable - For values around 1,000,000: 1e-3 may be needed - For values that went through many operations: use larger tolerances

0.2.27 Logical Operators

```
bool x = true, y = false;

bool and_result = x && y;    // false (both must be true) -- Python: x and y
bool or_result  = x || y;    // true  (either must be true) -- Python: x or y
bool not_result = !x;        // false (flip)                -- Python: not x
```

0.2.27.1 Short-Circuit Evaluation

&& stops evaluating as soon as it finds a false. || stops as soon as it finds a true. This is identical to Python and is critically important for safety:

```

int* p = nullptr;

// Safe: if p is null, the right side is NEVER evaluated
if (p != nullptr && *p > 0) {
    std::cout << "positive pointee\n";
}
// Without short-circuit: *p when p is null = crash

// Common optimization: cheap check first, expensive check second
if (is_in_cache(key) || compute_and_cache(key)) {
    // compute_and_cache only called when not in cache
}

```

Short-circuit evaluation is relied upon for correctness, not just optimization.

0.2.28 Bitwise Operators

These operate on individual bits within integers. They appear constantly in systems programming, graphics, embedded code, and game engines.

Op	Name	Meaning
&	AND	Output bit is 1 only if BOTH input bits are 1
\	OR	Output bit is 1 if EITHER input bit is 1
^	XOR	Output bit is 1 if the two input bits are DIFFERENT
~	NOT	Flips all bits
<<	Left shift	Shifts bits left, filling with 0 on the right
>>	Right shift	Shifts bits right (fills with 0 for unsigned, sign bit for signed)

Example: a = 5 = 0101 in binary
b = 3 = 0011 in binary

```

a & b:  0101
        0011
        ----
        0001   = 1   (only bit 0 is set in both)

```

```

a | b:  0101
        0011
        ----
        0111   = 7   (any bit set in either)

```

```

a ^ b:  0101
        0011
        ----
        0110  = 6  (bits that differ)

```

```
~a:      ~0101 = 1111...11111010 = -6 (for 32-bit signed int)
```

Left shift by 1 = multiply by 2. Left shift by n = multiply by 2^n :

```

1 << 0 = 1
1 << 1 = 2
1 << 2 = 4
1 << 3 = 8
1 << 4 = 16
1 << 10 = 1024

```

0.2.28.1 Practical Use: Boolean Flags Packed in an Integer

Instead of a separate `bool` for each property, pack them into one integer – each bit represents one flag:

```

const uint32_t VISIBLE    = 1u << 0;  // bit 0: 00000001
const uint32_t SOLID     = 1u << 1;  // bit 1: 00000010
const uint32_t ACTIVE    = 1u << 2;  // bit 2: 00000100
const uint32_t HAS_HEALTH = 1u << 3;  // bit 3: 00001000

uint32_t entity_flags = 0;

// Set a flag (turn bit ON):
entity_flags |= VISIBLE;  // flags = 00000001
entity_flags |= ACTIVE;   // flags = 00000101

// Test a flag (check if bit is set):
if (entity_flags & VISIBLE) { // nonzero = true
    render(entity);
}
if (entity_flags & ACTIVE) {
    update(entity);
}

// Clear a flag (turn bit OFF):
entity_flags &= ~ACTIVE;  // ~ACTIVE = 11111011; AND clears bit 2
                        // flags = 00000001

// Toggle a flag (flip):
entity_flags ^= SOLID;    // XOR flips bit 1

```

This pattern is everywhere: OpenGL uses `GL_COLOR_BUFFER_BIT | GL_DEPTH_BUFFER_BIT`, Vulkan uses `VkImageUsageFlags`, Linux uses `O_RDONLY | O_CREAT` for file open flags.

0.2.29 Operator Precedence

When operators are mixed, precedence determines the order of evaluation:

Highest priority (evaluated first)

<code>() [] -> .</code>	grouping, subscript, member access
<code>! ~ ++x --x (unary)</code>	unary operators (right-to-left)
<code>* / %</code>	multiplicative
<code>+ -</code>	additive
<code><< >></code>	bitwise shift
<code>< <= > >=</code>	relational comparison
<code>== !=</code>	equality comparison
<code>&</code>	bitwise AND
<code>^</code>	bitwise XOR
<code> </code>	bitwise OR
<code>&&</code>	logical AND
<code> </code>	logical OR
<code>?:</code>	ternary (right-to-left)
<code>= += -= ...</code>	assignment (right-to-left)

Lowest priority (evaluated last)

Common traps:

```
// Trap 1: bitwise AND vs equality
if (flags & MASK == 1) { }    // parsed as: flags & (MASK == 1)  -- WRONG
if ((flags & MASK) == 1) { }  // correct

// Trap 2: shift and arithmetic
int x = 2 + 3 << 1;    // parsed as: 2 + (3 << 1) = 2 + 6 = 8
int y = (2 + 3) << 1;  // = 5 << 1 = 10

// Rule: when in doubt, parenthesize
```

0.2.30 The Ternary Operator

```
# Python
label = "even" if n % 2 == 0 else "odd"
```

```
// C++
std::string label = (n % 2 == 0) ? "even" : "odd";
//                condition   true    false
```

The ternary is an *expression* that produces a value. Good for simple one-liners. Do not nest them – `a ? b ? c : d : e` is unreadable.

0.2.31 Common Mistakes in This Chapter

0.2.31.1 Mistake 1: Assignment Instead of Comparison

The bug:

```
if (x = 5) { ... } // assigns 5 to x, condition is always true (5 != 0)
```

The symptom: The `if` branch always runs; the `else` never runs; `x` silently changed. **The fix:** `if (x == 5)`. Compile with `-Wall` to get a warning about this. **Defensive coding:** Some people write `if (5 == x)` (“Yoda conditions”). Then a typo `if (5 = x)` is a compile error (can’t assign to a literal).

0.2.31.2 Mistake 2: Integer Division Surprise

The bug:

```
double bmi = weight / (height * height); // all ints -- integer division!
```

The fix: Cast one operand: `static_cast<double>(weight) / (height * height)`

0.2.31.3 Mistake 3: == on Floats

The bug: `if (result == expected_value)` where both are doubles **The fix:** `if (std::abs(result - expected_value) < 1e-9)`

0.2.31.4 Mistake 4: Bitwise vs Logical Operators

The bug:

```
bool a = is_ready();
bool b = can_proceed();
if (a & b) { ... } // bitwise AND -- works only if a and b are 0 or 1
// breaks for general integers
if (a && b) { ... } // logical AND -- correct for boolean conditions
```

The fix: Use `&&` and `||` for boolean logic. Use `&` and `|` for bit manipulation.

0.2.32 Exercises

Exercise 3.1 – Predict the output

```
int a = 17, b = 5;
std::cout << a / b << "\n";
std::cout << a % b << "\n";
std::cout << -a / b << "\n";
std::cout << -a % b << "\n";
```

Answer: 3, 2, -3, -2. C++ truncates toward zero; the sign of % follows the dividend.

Exercise 3.2 – Bit flag operations

Starting from `uint8_t flags = 0b00000000`: 1. Set bit 2 (value 4) 2. Set bit 5 (value 32) 3. Check if bit 2 is set (should be true) 4. Clear bit 2

Write the four operations and show the final value of `flags`.

Answer:

```
uint8_t flags = 0b00000000;

flags |= (1u << 2);           // 1. set bit 2: flags = 0b00000100
flags |= (1u << 5);           // 2. set bit 5: flags = 0b00100100

bool bit2_set = (flags & (1u << 2)) != 0; // 3. check bit 2: true

flags &= ~(1u << 2);           // 4. clear bit 2: flags = 0b00100000
// ~(1u << 2) = ~0b00000100 = 0b11111011
// 0b00100100 & 0b11111011 = 0b00100000 = 32
```

Exercise 3.3 – Floating-point tolerance

Write a function `are_equal(double a, double b)` that returns true when `a` and `b` are within 0.0001 of each other. Test it with `0.1 + 0.2` vs `0.3`.

Answer:

```
#include <cmath>
bool are_equal(double a, double b) {
    return std::abs(a - b) < 0.0001;
}
// are_equal(0.1 + 0.2, 0.3) returns true
// are_equal(0.1 + 0.2, 0.4) returns false
```

0.2.33 if/else if/else

```
# Python
score = 85
if score >= 90:
    grade = "A"
elif score >= 80:
    grade = "B"
else:
    grade = "C"
```

```
// C++
int score = 85;
std::string grade;

if (score >= 90) {
    grade = "A";
} else if (score >= 80) {
    grade = "B";
} else {
    grade = "C";
}
```

Three syntax differences: 1. The condition must be in **parentheses**: `if (score >= 90)` not `if score >= 90` 2. The body uses **curly braces** `{}` instead of `:` and indentation 3. **else if** (two words, with a space) not `elif`

0.2.33.1 Always Use Curly Braces – A Real Cautionary Tale

Braces are technically optional when the body is a single statement:

```
if (error)
    std::cout << "Error occurred\n";    // technically fine
```

But do not do this. The Apple “goto fail” bug (2014) used this pattern and allowed attackers to bypass SSL certificate verification on millions of Apple devices. The vulnerable code looked like:

```
if ((err = SSLHashSHA1.update(&hashCtx, &signedParams)) != 0)
    goto fail;
goto fail;    // This ALWAYS executes -- attacker bypasses verification
```

The second `goto fail` is not inside the `if`. The indentation lies. Always use braces.

0.2.33.2 if with Initializer (C++17)

Declare a variable scoped only to the `if/else` block:

```
// C++ before 2017:
int value = compute();
if (value > 0) {
    use(value);
}
// 'value' still exists here (wasted scope, potential confusion)

// C++17 and later:
if (int value = compute(); value > 0) {
    // ^-- initializer      ^-- condition
    use(value);
} else {
    handle_error(value);
}
// 'value' does not exist here -- scope is clean
```

Use this when the variable is only meaningful within the `if/else`.

0.2.34 switch

`switch` tests one integer or enum value against a list of constants:

```
int day = 3;

switch (day) {
    case 1:
        std::cout << "Monday\n";
        break;
    case 2:
        std::cout << "Tuesday\n";
        break;
    case 3:
        std::cout << "Wednesday\n";
        break;
    case 6:
    case 7:
        std::cout << "Weekend\n";    // cases 6 and 7 share the same body
        break;
    default:
        std::cout << "Weekday\n";
        break;
}
```

0.2.34.1 Fallthrough: The break Requirement

Without `break`, execution falls through into the next case:


```
int x = 1;
switch (x) {
    case 1:
        std::cout << "one\n";
        // no break!
    case 2:
        std::cout << "two\n";
        break;
    case 3:
        std::cout << "three\n";
        break;
}
```

Output when `x == 1`: one then two. The control falls through from case 1 into case 2.

This is almost always a bug. Always add `break`. If fallthrough is intentional, use `[[fallthrough]]` (C++17) to document it and silence the warning:

```
case 'A':
    [[fallthrough]]; // intentional: A and a are treated the same
case 'a':
    process_letter_a();
    break;
```

`switch` only works on integral types (`int`, `char`, `enum`). It does not work on strings or floating-point numbers.

0.2.35 while Loop

Checks the condition before each iteration. If the condition is false initially, the body never runs.

```
# Python
x = 0
while x < 5:
    print(x)
    x += 1
```

```
// C++
int x = 0;
while (x < 5) {
    std::cout << x << "\n";
    ++x;
}
// prints 0, 1, 2, 3, 4
```

0.2.36 do-while Loop

The body runs at least once. The condition is checked after the body.

```
int x = 100;
do {
    std::cout << x << "\n"; // prints 100 even though 100 >= 5
    ++x;
} while (x < 5);
// body ran once; condition was false; loop ends
```

When to use it: Input validation – you always want to ask once, then check.

```
#include <iostream>

int main() {
    int age;
    do {
        std::cout << "Enter your age (1-120): ";
        std::cin >> age;
    } while (age < 1 || age > 120);
    // Guaranteed: asked at least once; keeps asking until valid input

    std::cout << "Age accepted: " << age << "\n";
    return 0;
}
```

Python has no do-while. The equivalent Python idiom:

```
while True:
    age = int(input("Enter your age (1-120): "))
    if 1 <= age <= 120:
        break
```

0.2.37 for Loop: C-Style

```
for (initializer; condition; increment) {
    body;
}
```

```
for (int i = 0; i < 10; ++i) {
    std::cout << i << " "; // 0 1 2 3 4 5 6 7 8 9
}
// 'i' does not exist after the loop -- scoped to the for
```

Execution order:

```

for (int i = 0; i < 10; ++i)

    int i = 0      <-- runs ONCE before anything
        |
        v
    [check: i < 10?]  ----NO----> exit loop
        |
        YES
        |
        v
    [body: cout << i]
        |
        v
    [++i]
        |
        +-----> [check: i < 10?] (repeat)

```

Key point: the variable declared in the initializer (`int i`) exists only inside the `for` loop. This is better than Python, where loop variables leak out:

```

# Python: loop variable persists after loop
for i in range(5):
    pass
print(i)  # 4 -- still accessible

```

```

// C++: loop variable is gone after loop
for (int i = 0; i < 5; ++i) { }
std::cout << i;  // COMPILE ERROR: 'i' was not declared in this scope

```

Useful patterns:

```

// Count down:
for (int i = 10; i >= 0; --i) {
    std::cout << i << " ";  // 10 9 8 7 6 5 4 3 2 1 0
}

// Step by 3 (like range(0, 30, 3)):
for (int i = 0; i < 30; i += 3) {
    std::cout << i << " ";  // 0 3 6 9 12 15 18 21 24 27
}

// Multiple variables:
for (int i = 0, j = 10; i < j; ++i, --j) {
    std::cout << i << " " << j << "\n";
}
// 0 10, 1 9, 2 8, 3 7, 4 6

```

0.2.38 Range-Based for Loop (C++11)

The C++ equivalent of Python's `for x in collection`:

```
numbers = [10, 20, 30, 40, 50]
for n in numbers:
    print(n)
```

```
#include <vector>
std::vector<int> numbers = {10, 20, 30, 40, 50};
for (int n : numbers) {
    std::cout << n << "\n";
}
```

0.2.38.1 The Reference vs Copy Decision

This is the most important thing to understand about range-based for in C++.

By value (copy): Each iteration creates a copy. Modifying the loop variable does NOT change the container.

```
std::vector<int> v = {1, 2, 3};
for (int n : v) {
    n *= 10;    // modifies the copy, not the element in v
}
// v is still {1, 2, 3}
```

By reference (&): The loop variable IS the element. Modifying it DOES change the container.

```
for (int& n : v) {
    n *= 10;    // modifies the actual element in v
}
// v is now {10, 20, 30}
```

By const reference (const&): Read-only access to the element, no copy made. Use this for large objects you only need to read.

```
for (const int& n : v) {
    std::cout << n << "\n";    // read-only, no copy
    // n = 0;    // COMPILE ERROR: n is const
}
```

When to use which:

Type is small (int, double, bool, char, pointer):

```
for (T x : v)            -- copy is cheap, fine
for (const T& x : v)     -- also fine
```

Type is large (string, struct, vector, custom class):

```
for (const T& x : v)    -- ALWAYS prefer: no copy, no modify
for (T& x : v)         -- when you need to modify in place
```

You don't know the type or it's complex:

```
for (const auto& x : v) -- safe universal choice
```

0.2.39 break and continue

```
for (int i = 0; i < 10; ++i) {
    if (i == 5) {
        break;           // exit the loop immediately
    }
    if (i % 2 == 0) {
        continue;       // skip to the next iteration
    }
    std::cout << i << " "; // prints: 1 3
}
// loop ends because of break when i == 5
```

break and continue only affect the **innermost** enclosing loop. For nested loops, use a function return or a flag:

```
// Method: refactor into a function, use return
bool find_target(const std::vector<std::vector<int>>& grid, int target) {
    for (int row = 0; row < (int)grid.size(); ++row) {
        for (int col = 0; col < (int)grid[row].size(); ++col) {
            if (grid[row][col] == target) {
                return true; // exits both loops
            }
        }
    }
    return false;
}
```

0.2.40 Loop Execution Visualization

For `for (int i = 0; i < 3; ++i) { body; }:`

Start:

```
int i = 0    (initialization -- runs once)
```

Iteration 1:

```
i < 3? YES (0 < 3)
body executes
++i -> i = 1
```

Iteration 2:

```
i < 3? YES (1 < 3)
body executes
++i -> i = 2
```

Iteration 3:

```
i < 3? YES (2 < 3)
body executes
++i -> i = 3
```

Check:

```
i < 3? NO (3 is not < 3)
Loop exits
```

0.2.41 Common Mistakes in This Chapter

0.2.41.1 Mistake 1: Off-By-One Error

The bug:

```
std::vector<int> v = {10, 20, 30};
for (int i = 0; i <= v.size(); ++i) { // <= instead of <
    std::cout << v[i] << "\n";      // accesses v[3] -- does not exist!
}
```

The symptom: Crash, garbage output, or silent memory corruption on the last iteration.

The fix: Use `i < v.size()` (strictly less than). Or better: use range-based `for`, which cannot have this bug.

0.2.41.2 Mistake 2: Forgetting `break` in `switch`

The bug:

```
switch (command) {
    case 'q':
        prepare_to_quit();
        // forgot break -- falls through to 'h'!
    case 'h':
        show_help();
        break;
}
// When command == 'q': prepare_to_quit() runs, THEN show_help() runs
```

The fix: Add break after every case.

0.2.41.3 Mistake 3: Modifying a Container While Iterating Over It

The bug:

```
std::vector<int> v = {1, 2, 3, 4, 5};
for (int n : v) {
    if (n == 3) v.erase(v.begin() + 2); // modifies v while iterating!
}
// undefined behavior -- the range-based for's internal iterator is now invalid
```

The fix: Collect things to remove first, then remove them after the loop. Or use `std::remove_if` (Chapter 29).

0.2.41.4 Mistake 4: Using = Instead of == in a Condition

The bug:

```
int x = 5;
while (x = 10) { // assigns 10 to x, condition is always true (infinite loop!)
    std::cout << x << "\n";
}
```

The symptom: Infinite loop.

The fix: `while (x == 10)`. Compile with `-Wall` to get a warning.

0.2.42 Exercises

Exercise 4.1 – Predict the output

```
for (int i = 1; i <= 10; ++i) {  
    if (i % 3 == 0) continue;  
    if (i > 7) break;  
    std::cout << i << " ";  
}
```

Answer: 1 2 4 5 7 – skips multiples of 3 (3, 6), breaks when $i > 7$ (stops before 8).

Exercise 4.2 – FizzBuzz

Print numbers 1 to 20. Multiples of 3: print “Fizz”. Multiples of 5: print “Buzz”. Multiples of both 3 and 5: print “FizzBuzz”.

Answer:

```
#include <iostream>  
int main() {  
    for (int i = 1; i <= 20; ++i) {  
        if (i % 15 == 0) std::cout << "FizzBuzz\n";  
        else if (i % 3 == 0) std::cout << "Fizz\n";  
        else if (i % 5 == 0) std::cout << "Buzz\n";  
        else std::cout << i << "\n";  
    }  
}
```

Check % 15 first. If you check % 3 first, 15 would print “Fizz” instead of “FizzBuzz”.

Exercise 4.3 – Sum with while

Use a `while` loop to compute the sum of integers from 1 to 100.

Answer:

```
int sum = 0, i = 1;  
while (i <= 100) {  
    sum += i;  
    ++i;  
}  
std::cout << sum << "\n"; // 5050
```

Exercise 4.4 – Triangle

Print this triangle with nested loops:

```
*  
* *  
* * *  
* * * *  
* * * * *
```

Answer:

```
for (int row = 1; row <= 5; ++row) {  
    for (int col = 1; col <= row; ++col) {  
        std::cout << "* ";  
    }  
    std::cout << "\n";  
}
```

Exercise 4.5 – do-while validation

Write a program that asks the user to enter a password. Keep asking until they type “secret”. (Use `std::cin >> password` to read.)

Answer:

```
#include <iostream>  
#include <string>  
  
int main() {  
    std::string password;  
    do {  
        std::cout << "Enter password: ";  
        std::cin >> password;  
    } while (password != "secret");  
    std::cout << "Access granted.\n";  
    return 0;  
}
```

0.2.43 What Happens When You Call a Function

Before the syntax, understand what a function call does to memory. This mental model makes pass-by-value, scope, and the declaration/definition split all click at once.

Your program uses a region of memory called the **call stack**. It operates like a stack of plates: each function call pushes a new **frame** (plate) on top; when the function returns, that frame is popped off and gone.

Each frame holds: - The function's parameters (copies of the arguments) - The function's local variables - The return address (where to continue after returning)

```
int multiply(int a, int b) {
    int result = a * b;
    return result;
}

int square(int x) {
    int s = multiply(x, x);
    return s;
}

int main() {
    int answer = square(5);
    std::cout << answer << "\n"; // 25
}
```

When `square(5)` is called, then `multiply(5, 5)` is called inside `square`:

CALL STACK (top = currently executing function)

multiply's frame	<-- currently executing
a = 5	COPY of square's x
b = 5	COPY of square's x
result = 25	
square's frame	
x = 5	COPY of main's argument 5
s = ?	not yet assigned
main's frame	
answer = ?	not yet assigned

When `multiply` finishes: 1. It returns 25 2. Its entire frame is popped – `a`, `b`, `result` are gone from memory 3. Back in `square`, `s` receives the returned 25 4. `square` returns 25, its frame is popped – `x`, `s` are gone 5. In `main`, `answer` receives 25

Critical insight: `a` and `b` inside `multiply` are COPIES of `x` from `square`. They live at different memory addresses. Changing `a` inside `multiply` has absolutely no effect on `x` in `square`. This is **pass by value**.

0.2.44 Function Syntax

```
# Python -- no return type, no parameter types
def add(a, b):
    return a + b
```

```
// C++ -- return type required, parameter types required
int add(int a, int b) {
    return a + b;
}
```

What you must provide: 1. **Return type** before the function name. `int` means the function returns an `int`. 2. **Type for each parameter**. Each parameter is `type name`. 3. A return statement if the return type is not `void`.

```
// Returns nothing:
void print_banner() {
    std::cout << "=====\n";
} // no return needed for void

// Returns bool:
bool is_palindrome(std::string s) {
    std::string rev = s;
    std::reverse(rev.begin(), rev.end());
    return s == rev;
}

// Multiple parameters:
double distance(double x1, double y1, double x2, double y2) {
    double dx = x2 - x1;
    double dy = y2 - y1;
    return std::sqrt(dx*dx + dy*dy);
}

// No parameters:
int get_screen_width() {
    return 1920;
}
```

0.2.45 Declarations vs Definitions

This is a concept Python does not have. It exists because C++ compiles each `.cpp` file independently, top-to-bottom.

The problem: If you call a function before the compiler has seen it, the compiler does not know: - Does this function exist? - How many parameters does it take? - What types are the parameters? - What type does it return?

Without this information, the compiler cannot type-check the call. It refuses.

A declaration (also called a forward declaration or prototype) gives the compiler the function's signature without the body:

```
int add(int a, int b); // declaration: signature only, ends with semicolon
```

A definition is the full function with its body:

```
int add(int a, int b) { // definition: has the body in braces
    return a + b;
}
```

Every definition is implicitly also a declaration. A declaration without a body is not a definition.

0.2.45.1 The Three Solutions

Problem: function used before it is defined

```
int main() {
    int r = add(3, 4); // ERROR: 'add' not declared yet
}

int add(int a, int b) { return a + b; }
```

Solution 1: Move the definition above the call

```
int add(int a, int b) { return a + b; } // defined first

int main() {
    int r = add(3, 4); // OK -- compiler has seen add
}
```

Works for simple, single-file programs. Becomes unmanageable with circular dependencies (function A calls B, B calls A).

Solution 2: Forward declaration

```
int add(int a, int b); // declaration -- tells compiler the signature

int main() {
    int r = add(3, 4); // OK -- compiler knows signature, trusts you to define it
}

int add(int a, int b) { return a + b; } // definition can appear anywhere after
```

Works in a single file. For multi-file projects, use headers.

Solution 3: Header file (the real solution for multi-file projects)

```
// math.h -- declarations only
#pragma once
int add(int a, int b);
int subtract(int a, int b);

// math.cpp -- definitions
#include "math.h"
int add(int a, int b) { return a + b; }
int subtract(int a, int b) { return a - b; }

// main.cpp -- uses the functions
#include "math.h" // gives main.cpp the declarations it needs
int main() {
    int r = add(3, 4); // compiler checks types via the declaration in math.h
    int s = subtract(10, 3);
}
```

We cover headers fully in Chapter 11.

0.2.46 Pass by Value: What Is Copied and Why It Matters

When you pass a variable to a function, C++ copies it by default. The function gets its own independent copy.

```
void zero_it(int x) {
    x = 0; // modifies the local copy only
    std::cout << "Inside: " << x << "\n";
}

int main() {
    int n = 42;
    zero_it(n);
    std::cout << "Outside: " << n << "\n";
}
```

Output:

```
Inside: 0
Outside: 42    <-- n was NOT changed
```

The stack during the call:

```

main's frame:                                zero_it's frame:
+-----+                                +-----+
| n = 42      |                                | x = 42      | <-- a COPY at a different address
| (at 0x7fff1000) |                                | (at 0x7fff0ff0) |
+-----+                                +-----+
                                         |
                                         x = 0 (modifies copy)
                                         |
                                         zero_it returns, frame popped

main's frame:
+-----+
| n = 42      | <-- unchanged
+-----+

```

This applies to every type: `int`, `double`, `std::string`, structs, classes. Passing a `std::string` with 10,000 characters by value copies all 10,000 characters. This is why Chapter 6 (References) and `const&` parameters matter.

0.2.47 Default Parameters

```

# Python
def connect(host, port=8080, timeout=30):
    print(f"Connecting to {host}:{port} (timeout={timeout}s)")

```

```

// C++
void connect(std::string host, int port = 8080, int timeout = 30) {
    std::cout << "Connecting to " << host << ":" << port
              << " (timeout=" << timeout << "s)\n";
}

connect("localhost");           // port=8080, timeout=30
connect("localhost", 9000);     // port=9000, timeout=30
connect("localhost", 9000, 60); // port=9000, timeout=60

```

Rules: 1. Default parameters must be at the **end** of the parameter list. 2. You cannot skip a middle argument: `connect("host", , 60)` is illegal. 3. Specify defaults in the **declaration** (header file), not in the definition:

```

// math.h -- put defaults here
void connect(std::string host, int port = 8080, int timeout = 30);

// math.cpp -- no defaults here (would be a redefinition error)
void connect(std::string host, int port, int timeout) {
    // implementation
}

```

0.2.48 Function Overloading

C++ lets you define multiple functions with the same name, as long as they have different parameter types. The compiler picks the right one at each call site.

```
int    abs_val(int x)    { return x < 0 ? -x : x; }
double abs_val(double x) { return x < 0.0 ? -x : x; }
float  abs_val(float x)  { return x < 0.0f ? -x : x; }

abs_val(5);           // calls int version
abs_val(-3.14);       // calls double version
abs_val(-2.0f);       // calls float version (f suffix = float literal)
```

The compiler resolves overloads at compile time using these rules (simplified): 1. **Exact match** – preferred 2. **Trivial conversion** (e.g., non-const to const) 3. **Promotion** (e.g., float to double, short to int) 4. **Standard conversion** (e.g., int to double) 5. **Ambiguous** (two overloads equally good) – compile error

```
void foo(int x)    { std::cout << "int\n"; }
void foo(double x) { std::cout << "double\n"; }

foo(5);           // "int"    -- exact match
foo(5.0);         // "double" -- exact match
foo(5.0f);        // "double" -- float promotes to double (closer than int)
foo('A');         // "int"    -- char promotes to int
// foo(true);    -- ambiguous: bool can promote to int OR double equally well
```

Python does not have overloading. You achieve similar behavior with type checks (`isinstance`) or default parameters. The C++ approach is resolved entirely at compile time with zero runtime cost.

0.2.48.1 What Overloading Is NOT

Overloading is based on **parameter types only**. You cannot overload on return type:

```
int    get() { return 5; }
double get() { return 5.0; } // ERROR: same parameters, only return type differs
```

The compiler uses the call arguments to pick the overload. If two functions take the same arguments, there is no way to distinguish them.

0.2.49 Compiler Messages for Function Errors

Understanding compiler error messages makes debugging much faster. Here are the most common function-related errors:

0.2.49.1 No Matching Function

```
void greet(std::string name) { }
```

```
int main() {  
    greet(42);    // passing int, but greet takes string  
}
```

```
error: no matching function for call to 'greet(int)'  
note: candidate: 'void greet(std::string)'  
note:   no known conversion from 'int' to 'std::string'
```

The compiler tells you what it found and why it didn't match.

0.2.49.2 Ambiguous Overload

```
void process(int x) { }  
void process(long x) { }
```

```
process(42);    // 42 is int -- exact match for int version, fine  
process(42L);   // 42L is long -- exact match for long version, fine  
// process(true); -- ambiguous: bool can convert to int OR long
```

```
error: call to 'process' is ambiguous  
note: candidate: 'void process(int)'  
note: candidate: 'void process(long)'
```

0.2.49.3 Wrong Number of Arguments

```
void foo(int a, int b) { }
```

```
foo(1, 2, 3);    // too many
```

```
error: too many arguments to function 'void foo(int, int)'  
note: declared here: void foo(int a, int b)
```


0.2.50 Common Mistakes in This Chapter

0.2.50.1 Mistake 1: Expecting Pass-by-Value to Modify the Original

The bug:

```
void set_to_zero(int x) { x = 0; }

int count = 100;
set_to_zero(count);
std::cout << count;    // 100 -- NOT 0
```

Why it fails: `x` is a copy. Setting `x = 0` changes only the copy.

The fix: Use a reference (Chapter 6): `void set_to_zero(int& x) { x = 0; }`

0.2.50.2 Mistake 2: Missing Return Statement

The bug:

```
int max(int a, int b) {
    if (a > b) return a;
    // forgot: return b;
}
```

The symptom: Compiler warning “control may reach end of non-void function.” Calling `max` when `a <= b` returns garbage.

The fix: All code paths must return a value.

0.2.50.3 Mistake 3: Defining a Function Twice

The bug:

```
int add(int a, int b) { return a + b; }
int add(int a, int b) { return a + b; } // second definition
```

error: redefinition of 'int add(int, int)'

The fix: Declarations can appear multiple times (they’re just type information). Definitions can appear exactly once.

0.2.50.4 Mistake 4: Forgetting to Compile the File Containing the Definition

The bug:

```
$ g++ -std=c++23 -o program main.cpp # forgot greet.cpp
```

undefined reference to `greet(std::string const&)'

The fix: Include all .cpp files that contain definitions you need:

```
$ g++ -std=c++23 -o program main.cpp greet.cpp
```

0.2.51 Exercises

Exercise 5.1 – Declaration or definition?

For each, say whether it is a declaration, a definition, or both:

```
double sqrt(double x); // (a)
double square(double x) { return x*x; } // (b)
void print(); // (c)
void print() { std::cout << "!\n"; } // (d)
```

Answer: - (a): declaration only (no body) - (b): definition (has body) – and also implicitly a declaration - (c): declaration only - (d): definition (and implicitly a declaration)

Exercise 5.2 – Stack trace

Draw the call stack when `inner` is executing:

```
int inner(int x) { return x * 2; }
int middle(int x) { return inner(x + 1); }
int outer(int x) { return middle(x + 1); }
int main() { int r = outer(5); }
```

Answer:

```
inner's frame:  x = 7    (outer 5 -> middle 6 -> inner 7)
middle's frame: x = 6
outer's frame:  x = 5
main's frame:   r = ?
```

inner returns $7 * 2 = 14$. Pops. middle returns 14. Pops. outer returns 14. Pops. main's $r = 14$.

Exercise 5.3 – Overloaded describe

Write three overloaded functions named `describe` that: - Take `int` and print "Integer: N" - Take `double` and print "Double: N.NN" - Take `std::string` and print "String: S (length L)"

Answer:

```
#include <iostream>
#include <string>
#include <iomanip>

void describe(int n) {
    std::cout << "Integer: " << n << "\n";
}

void describe(double d) {
    std::cout << "Double: " << std::fixed << std::setprecision(2) << d << "\n";
}

void describe(std::string s) {
    std::cout << "String: " << s << " (length " << s.size() << ")\n";
}

int main() {
    describe(42);
    describe(3.14159);
    describe(std::string{"hello"});
}
```

Output:

```
Integer: 42
Double: 3.14
String: hello (length 5)
```

Exercise 5.4 – power with default

Write `power (base, exponent=2)` that computes $\text{base}^{\text{exponent}}$ using a loop. It should work for non-negative integer exponents. Verify: `power (3)` returns 9, `power (2, 10)` returns 1024.

Answer:

```
double power(double base, int exponent = 2) {  
    double result = 1.0;  
    for (int i = 0; i < exponent; ++i) {  
        result *= base;  
    }  
    return result;  
}
```

Part I is complete. You now have the foundations: the compilation model, types and memory, operators, control flow, and functions.

Part II begins with the concepts that most distinguish C++ from Python: references, pointers, the stack vs the heap, and const correctness. These are where C++ becomes its own language. Ask to continue.

0.3 Part II – Core C++ (the part that is not like Python)

The next six chapters cover the concepts that have no real equivalent in Python. Python hides all of this behind its object model and garbage collector. C++ exposes it – and once you understand it, you understand why both languages make the choices they do.

Chapter 6: References – Aliases for Variables

0.3.1 The Problem References Solve

At the end of Chapter 5 you saw that passing a variable to a function copies it. That is usually fine for `int` and `double`. It is expensive for large objects and sometimes flat-out wrong when you need the function to modify the original.

```
// Costs nothing to copy:  
void print_count(int n) { std::cout << n << "\n"; }  
  
// Copying a 50,000-byte buffer is painful:  
void compress(HugeBuffer data) { ... } // copies 50,000 bytes on every call  
  
// Needs to modify the original -- copy is useless:  
void swap(int a, int b) {  
    int tmp = a;  
    a = b;      // modifies the copy, not main's variables  
    b = tmp;  
}  
  
int x = 10, y = 20;  
swap(x, y);  
// x is still 10, y is still 20 -- nothing swapped
```

References solve all three problems.

0.3.2 What Is a Reference?

A reference is an **alias** – another name for an existing variable. It is not a copy. It is not a pointer (though it is implemented as one). It is a second name that refers to the exact same memory location.

```
int x = 42;
int& ref = x;    // ref is a reference to x. & after the type = reference.

std::cout << x    << "\n";    // 42
std::cout << ref  << "\n";    // 42 -- same memory, same value

ref = 100;        // modifying ref modifies x (they ARE the same thing)
std::cout << x    << "\n";    // 100
std::cout << ref  << "\n";    // 100
```

Memory picture:

Before: `ref = x` declared

Address	Name	Value
0x1000	x	42

\

- +-- ref is another name for this same location
- | No new memory is allocated for ref itself.
- | (The compiler may use a pointer internally,
- | but you never see it.)

Rules for references: 1. A reference **must** be initialized when declared. `int& r;` is a compile error. 2. A reference **cannot be reseated** – once bound to a variable, it refers to that variable for its entire life. 3. After initialization, using `ref` is exactly like using the original variable.

```
int a = 1, b = 2;
int& r = a;    // r refers to a
r = b;        // this does NOT make r refer to b
              // it assigns b's value (2) TO a via r
std::cout << a; // 2
// r still refers to a, not b
```

0.3.3 Pass by Reference

To let a function modify its argument, pass it by reference:

```
void swap(int& a, int& b) {    // & in parameter = reference parameter
    int tmp = a;
    a = b;
    b = tmp;
}

int x = 10, y = 20;
swap(x, y);
std::cout << x << " " << y;    // 20 10 -- actually swapped
```

Inside swap, a IS x and b IS y. They share the same memory.

main's frame:		swap's frame:	
+-----+		+-----+	
x = 10	<--->	a (reference)	a and x ARE the same memory
y = 20	<--->	b (reference)	b and y ARE the same memory
+-----+		+-----+	
		a = 20, b = 10 executed	
+-----+			
x = 20		x changed through a	
y = 10		y changed through b	
+-----+			

Contrast with pass-by-value from Chapter 5:

```
void swap_broken(int a, int b) {    // no &, copies made
    int tmp = a; a = b; b = tmp;    // swaps the copies only
}

swap_broken(x, y);    // x and y unchanged
```

0.3.4 Pass by const Reference

When you want to avoid copying a large object but do NOT need to modify it, use const&:

```
// BAD: copies the entire string (potentially many bytes)
void print_name(std::string name) {
    std::cout << name << "\n";
}

// GOOD: no copy, but cannot modify (const ensures this)
```

```
void print_name(const std::string& name) {
    std::cout << name << "\n";
    // name = "Bob"; // COMPILER ERROR -- const reference, cannot assign
}
```

`const std::string& name` means: - & – reference (no copy) - const – cannot be modified through this reference

The caller's string is not copied. The function gets direct read-only access to the original. This is the most common parameter style for anything larger than a few bytes.

0.3.4.1 The Decision Table for Parameter Style

Parameter type	When to use
<code>int x</code>	Small types (int, double, bool, char, pointer) that are cheap to copy and you don't need to modify.
<code>int& x</code>	Small or large type that the function MUST modify. (signals to the caller: "I will change your variable")
<code>const std::string& x</code>	Large type (string, vector, struct) that you only need to read, not modify. No copy. Fastest.
<code>std::string& x</code>	Large type that the function must modify. (rare -- usually return a new value instead)

0.3.5 References Cannot Be Null

Unlike pointers (next chapter), a reference is guaranteed to refer to a valid object. You cannot have a “null reference” in well-formed C++.

```
int& bad_ref = *nullptr; // undefined behavior -- DO NOT DO THIS
                       // dereferencing null pointer gives "reference to nothing"
```

If validity is uncertain at the time of binding, use a pointer instead. References are for when you know the object exists.

0.3.6 Returning References

A function can return a reference to give the caller direct access to something inside the function... but only if that thing outlives the function call.

```
// CORRECT: returning a reference to a member of an object that outlives the call
int& get_element(std::vector<int>& v, int index) {
    return v[index];    // v exists outside this function; safe
}

std::vector<int> nums = {10, 20, 30};
get_element(nums, 1) = 99;    // assigns through the returned reference
// nums is now {10, 99, 30}
```

```
// WRONG: returning a reference to a local variable
int& bad() {
    int x = 5;
    return x;    // x is destroyed when bad() returns -- dangling reference!
}
int& r = bad();    // r refers to memory that no longer exists
r = 10;    // undefined behavior -- corrupts memory
```

The compiler often warns about this:

warning: reference to local variable 'x' returned [-Wreturn-local-addr]

```
7 |     return x;
  |           ^
```

0.3.7 Common Mistakes in This Chapter

0.3.7.1 Mistake 1: Expecting pass-by-reference from a non-reference parameter

The bug:

```
void double_it(int n) { n *= 2; }    // n is a copy
int x = 5;
double_it(x);
std::cout << x;    // 5, not 10
```

The fix: `void double_it(int& n) { n *= 2; }`

0.3.7.2 Mistake 2: Forgetting `const` on read-only reference parameters

The bug:


```
void print_sum(std::string& s) { // non-const reference
    std::cout << s << "\n";
}
print_sum("hello"); // COMPILE ERROR: cannot bind non-const reference to rvalue
```

Why it fails: Temporary values (like string literals) cannot bind to non-const references because the compiler cannot take their address in a meaningful way. **The fix:** `void print_sum(const std::string& s)`

0.3.7.3 Mistake 3: Returning a reference to a local variable

The bug: As shown above – the local variable is destroyed on return, leaving a dangling reference. **The fix:** Only return references to objects that outlive the function (parameters, class members, statics).

0.3.8 Exercises

Exercise 6.1 – Trace the output

```
void increment(int& x) { ++x; }

int a = 5;
int& b = a;
increment(b);
std::cout << a << " " << b;
```

Answer: 6 6. `b` is a reference to `a`, so they are the same variable. `increment(b)` increments `a` (via `b` via the parameter `x`). Both names show the new value.

Exercise 6.2 – Fix the swap

```
void swap(double a, double b) {
    double tmp = a;
    a = b;
    b = tmp;
}

double x = 1.5, y = 2.5;
swap(x, y);
// x and y are unchanged -- fix swap so they actually swap
```

Answer:

```
void swap(double& a, double& b) {
    double tmp = a;
    a = b;
    b = tmp;
}
```

Exercise 6.3 – const& parameter

Write a function `print_stats(const std::vector<int>& v)` that prints the size, first element, and last element of a vector. It should not copy the vector.

Answer:

```
#include <iostream>
#include <vector>

void print_stats(const std::vector<int>& v) {
    std::cout << "Size: " << v.size() << "\n";
    std::cout << "First: " << v.front() << "\n";
    std::cout << "Last: " << v.back() << "\n";
}

int main() {
    std::vector<int> nums = {10, 20, 30, 40, 50};
    print_stats(nums); // no copy of nums made
}
```

Chapter 7: Pointers and Memory Addresses

0.3.9 The Address of a Variable

Every variable lives at a specific location in memory – a numbered address. On a 64-bit system, addresses are 64-bit numbers, usually displayed in hexadecimal.

```
int x = 42;
std::cout << &x << "\n"; // prints something like: 0x7ffee4b3c5ac
```

The `&` operator (when used in an expression, not a declaration) is the **address-of** operator. It gives you the memory address where a variable lives.

Variable `x`:

Address	Value
0x7fff10	[00][00][00][2A] <- &x is 0x7fff10; *(&x) is 42

A **pointer** is a variable that stores an address. The `*` in the type means “pointer to”:

```
int x = 42;           // x is an int
int* p = &x;          // p is a pointer to int, stores x's address

std::cout << p        << "\n";    // 0x7fff10 -- the address (the pointer value)
std::cout << *p        << "\n";    // 42      -- the value at that address (dereference)
std::cout << &p        << "\n";    // 0x7fff08 -- address of the pointer itself
```

The `*` when used with an existing pointer is the **dereference** operator – it follows the address to get the value stored there.

Memory picture:

Address	Variable	Value	
0x7fff10	x (int)	42	
0x7fff08	p (int*)	0x7fff10	<- p stores x's address

p is the address 0x7fff10.
*p follows that address and gives 42.

0.3.10 Modifying Through a Pointer

```
int x = 42;
int* p = &x;

*p = 100; // write 100 to the memory location p points to
           // same as: x = 100

std::cout << x;    // 100
std::cout << *p;   // 100
```

This is the lower-level version of references. Pointers and references both allow indirect modification, but:

	Reference	Pointer
Syntax	int& r = x	int* p = &x
Must be initialized	Yes	No (but you should)
Can be null	No	Yes (nullptr)

	Reference	Pointer
Can be reseated	No	Yes
Dereference syntax	just use r	*p
When to use	Aliases, function parameters	Optional relationships, arrays, dynamic memory

0.3.11 nullptr: The Null Pointer

A pointer that points to nothing is called a null pointer. Always use `nullptr` (C++11), never `NULL` (old C macro) or `0`:

```
int* p = nullptr; // p points to nothing

if (p != nullptr) {
    std::cout << *p; // safe -- only dereference if not null
}

// Dereferencing a null pointer is undefined behavior (usually a crash):
int* q = nullptr;
std::cout << *q; // CRASH -- segmentation fault
```

Null pointer:

Address	Variable	Value
0x7fff08	p (int*)	0x0000000000000000 <- nullptr (address zero)

Dereferencing: `*p` means "go to address 0 and read from there"
OS protects address 0 -- your program gets SIGSEGV (segfault)

Always check pointers before dereferencing if they might be null.

0.3.12 Pointer Arithmetic

Pointers support arithmetic that advances by the size of the pointed-to type:

```
int arr[5] = {10, 20, 30, 40, 50};
int* p = arr;    // p points to arr[0]

std::cout << *p      << "\n";    // 10
std::cout << *(p + 1) << "\n";    // 20 (moves 4 bytes forward, one int)
std::cout << *(p + 2) << "\n";    // 30

p++;             // p now points to arr[1]
std::cout << *p      << "\n";    // 20
```

Memory (each int is 4 bytes):

Address	Value	
0x1000	10	<- arr[0], p points here initially
0x1004	20	<- arr[1], p+1 points here
0x1008	30	<- arr[2]
0x100C	40	<- arr[3]
0x1010	50	<- arr[4]

$p + 1$ moves by `sizeof(int) = 4` bytes, landing on the next integer. This is why arrays are contiguous in memory – pointer arithmetic only works correctly because of that contiguity.

0.3.13 Pointers vs Python References

In Python, every variable is a reference (pointer) to an object. Python programmers are already using pointers – they just don't see the mechanics.

```
# Python: all "variables" are pointers
a = [1, 2, 3]
b = a    # b and a point to the SAME list
b.append(4)
print(a)    # [1, 2, 3, 4] -- a was affected through b

# Python explicitly shows this with id():
print(id(a) == id(b))    # True -- same memory address
```

```
// C++: copying by default, pointer explicitly
std::vector<int> a = {1, 2, 3};
std::vector<int> b = a;    // b is a COPY -- different memory
b.push_back(4);
std::cout << a.size();    // 3 -- a is unaffected

std::vector<int>* p = &a;    // p points to a
p->push_back(4);    // modifies a through the pointer
std::cout << a.size();    // 4
```

In Python, `b = a` for a list makes two labels for one object. In C++, `b = a` copies everything. To share, you explicitly use a pointer or reference.

0.3.14 The `->` Operator

When you have a pointer to an object and want to access its members, `->` is shorthand for `(*p).member`:

```
struct Point {
    double x, y;
};

Point pt{3.0, 4.0};
Point* p = &pt;

std::cout << (*p).x << "\n"; // 3.0 -- dereference then member access
std::cout << p->x << "\n";   // 3.0 -- same thing, cleaner syntax
p->x = 10.0;                  // modifies pt.x through the pointer
```

`->` is the idiomatic way to access members through a pointer. You will see it constantly in C++ code.

0.3.15 `const` and Pointers

There are two things that can be `const` with a pointer: the pointer itself (where it points) and the value it points to.

```
int a = 1, b = 2;

// Pointer to const int: cannot change the pointed-to value
const int* p1 = &a;
*p1 = 10;      // ERROR: *p1 is const
p1 = &b;       // OK: p1 itself can be changed (points to b now)

// Const pointer to int: cannot change where it points
int* const p2 = &a;
*p2 = 10;      // OK: the int it points to can be changed
p2 = &b;       // ERROR: p2 is const (cannot reset)

// Const pointer to const int: cannot change either
const int* const p3 = &a;
*p3 = 10;      // ERROR
p3 = &b;       // ERROR

// Memory trick: read the type right-to-left
// const int*      --> pointer to const int
// int* const      --> const pointer to int
// const int* const --> const pointer to const int
```

0.3.16 Common Mistakes in This Chapter

0.3.16.1 Mistake 1: Dereferencing a Null or Uninitialized Pointer

The bug:

```
int* p;           // uninitialized -- contains garbage address
*p = 5;           // writes to garbage address -- undefined behavior
```

The symptom: Immediate crash (segfault), or silent memory corruption that crashes later.

The fix: Always initialize pointers: `int* p = nullptr;` or `int* p = &some_variable;`

0.3.16.2 Mistake 2: Using a Pointer After the Pointed-To Variable is Destroyed

The bug:

```
int* get_ptr() {
    int local = 42;
    return &local; // local is destroyed when function returns
}
int* p = get_ptr(); // p now points to freed stack memory
std::cout << *p;    // undefined behavior -- dangling pointer
```

The fix: Never return a pointer to a local variable. Return the value, or make the variable static, or allocate on the heap (Chapter 8).

0.3.16.3 Mistake 3: Confusing & as Address-Of vs & as Reference Type

The confusion:

```
int x = 5;
int& r = x; // & in declaration = reference type
int* p = &x; // & in expression = address-of operator
```

The position matters: & after a type in a declaration means “reference.” & before a variable in an expression means “take the address of.”

0.3.17 Exercises

Exercise 7.1 – Trace the pointer

```
int a = 10, b = 20;
int* p = &a;
*p = 30;
p = &b;
*p += 5;
std::cout << a << " " << b;
```

Answer: 30 25. Step by step: `*p = 30` sets a to 30. `p = &b` makes p point to b. `*p += 5` adds 5 to b (20 + 5 = 25).

Exercise 7.2 – Pointer parameter

Rewrite `double_it` to take a pointer parameter instead of a reference:

```
void double_it(int* p) {
    *p *= 2;
}
int x = 7;
double_it(&x);    // must pass address explicitly
std::cout << x;   // 14
```

Which style (reference or pointer) is more idiomatic in modern C++ for this use case? Why?

Answer: Reference is more idiomatic. It is simpler (`double_it(x)` vs `double_it(&x)`), cannot be null, and cannot be reseated. Pointers are used when nullability or reseating is needed.

Exercise 7.3 – const correctness with pointers

Declare a pointer p to `double` such that: - The value at p cannot be changed through p - p itself can be pointed at a different `double`

Then declare q such that: - q is fixed to always point to the same `double` - The value at q CAN be changed through q

Answer:

```
double a = 1.0, b = 2.0;
const double* p = &a;    // ptr to const double
p = &b;                  // OK -- p can be reseated
// *p = 5.0;             // ERROR -- cannot modify through p

double* const q = &a;    // const pointer to double
*q = 5.0;                // OK -- can modify value
// q = &b;               // ERROR -- cannot reseate q
```

Chapter 8: The Stack and the Heap

0.3.18 Two Regions of Memory

Every running program has two main regions where variables can live:

+-----+	
Stack	
- Fast (just moves a pointer)	
- Automatic (compiler manages lifetime)	
- Limited size (~1-8 MB)	
- Local variables, function parameters	
+-----+	
(other memory: code, globals...)	
+-----+	
Heap	
- Slower (OS/allocator involved)	
- Manual (YOU manage lifetime in C++)	
- Huge (limited by RAM)	
- Dynamic allocation: new, malloc	
+-----+	

Python hides this entirely – all objects live on the heap, managed by the garbage collector. C++ exposes both regions and lets you choose where memory comes from.

0.3.19 The Stack in Detail

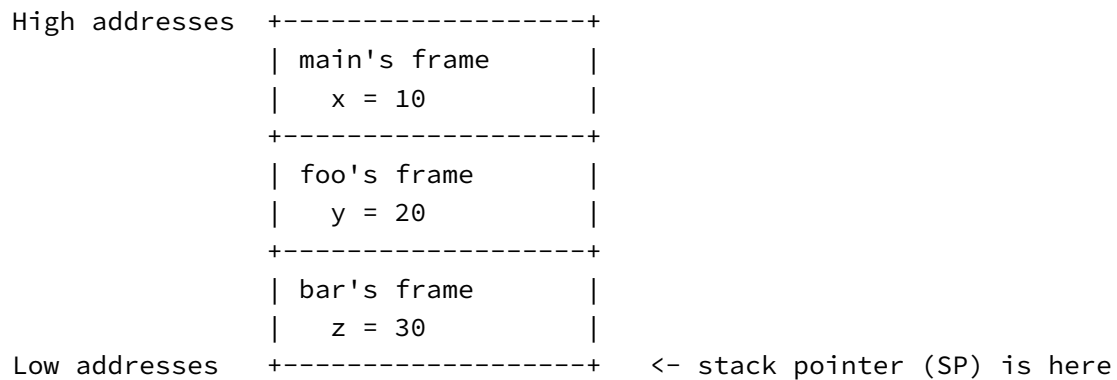
The stack is a LIFO (Last In, First Out) structure. Local variables and function parameters live here.

```
void bar() {
    int z = 30;
    // z lives on the stack while bar() is running
}

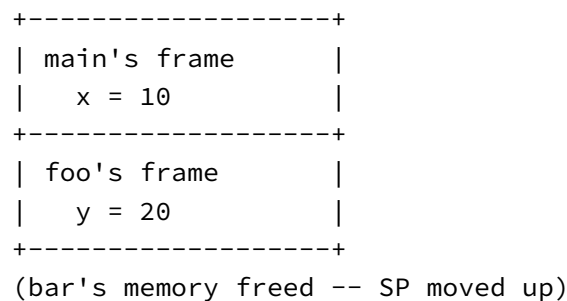
void foo() {
    int y = 20;
    bar();    // bar's frame pushed on top of foo's
    // z is gone (bar's frame popped); y is back on top
}

int main() {
    int x = 10;
    foo();
    // y is gone; x is still here
}
```

Call stack during bar():



After bar() returns:



Allocating stack memory is nearly free: the CPU just decrements the stack pointer register by the number of bytes needed. Freeing stack memory is also free: the CPU increments the stack pointer back.

Stack overflow: If you allocate too much on the stack (e.g., declaring a 10MB array locally, or infinitely deep recursion), the stack runs into other memory and the OS kills your program.

```
// Stack overflow example:
void recurse() { recurse(); } // infinite recursion
recurse(); // segfault when stack space exhausted

// Stack overflow with large local array:
void dangerous() {
    int huge[2000000]; // ~8 MB on the stack -- likely crashes
}
```

0.3.20 The Heap in Detail

The heap is a large pool of memory managed by the OS and your allocator. You request chunks with `new` and release them with `delete`.

```
int* p = new int{42};           // allocates 4 bytes on the heap; p stores the address
std::cout << *p << "\n";      // 42
delete p;                       // releases the 4 bytes back to the allocator
p = nullptr;                   // good practice: prevents accidental reuse
```

After `int* p = new int{42}`:

Stack:		Heap:
+-----+		+-----+
p = 0x555f2a1b10	----->	42 (4 bytes)
+-----+		+-----+

After `delete p`:

Stack:		Heap:
+-----+		(memory returned to heap pool)
p = 0x555f2a1b10	---X-->	(no longer valid to access)
+-----+		

After `p = nullptr`:

Stack:		
+-----+		
p = 0x0		(safe -- cannot accidentally dereference)
+-----+		

0.3.20.1 Heap Arrays

```
int n = 1000;
int* arr = new int[n]{};           // allocate n ints on the heap, zero-initialized
arr[0] = 10;
arr[n-1] = 99;

delete[] arr; // MUST use delete[] for arrays (not delete)
arr = nullptr;
```

The size does not need to be a compile-time constant – you can compute it at runtime. This is the main use case for heap allocation.

0.3.21 Why Python Uses the Heap for Everything

Python allocates every object on the heap and uses reference counting to track when objects can be freed. This is why Python is flexible but slower:

```
x = [1, 2, 3]    # list allocated on heap
y = x           # y is another reference to the same heap object
                # reference count goes from 1 to 2
del x           # reference count goes from 2 to 1 -- NOT freed
del y           # reference count goes from 1 to 0 -- freed!
```

C++ gives you the choice: - Stack: fast, automatic, size known at compile time - Heap: flexible, manual (or smart pointers), any size

```
// Stack -- automatic lifetime, fast
std::vector<int> v = {1, 2, 3}; // v is on the stack
// vector's CONTENTS are on the heap (internally), but you don't manage that
// v is automatically destroyed when it goes out of scope

// Heap -- manual lifetime (rare in modern C++ -- use smart pointers instead)
std::vector<int>* vp = new std::vector<int>{1, 2, 3};
delete vp; // you must remember to do this
```

0.3.22 Memory Leaks

A memory leak is heap memory that was allocated but never freed. In C++, the OS reclaims all memory when the process exits, so short-lived programs don't suffer from leaks permanently. But long-running programs (servers, games) slowly consume more and more memory until they crash or the machine runs out.

```
void leaky() {
    int* p = new int{42}; // allocates heap memory
    // forgot: delete p;
    // function returns -- p (the pointer) is gone
    // the heap memory at p's address is still allocated, now unreachable
}

for (int i = 0; i < 1000000; ++i)
    leaky(); // leaks 4 bytes per call = 4 MB leaked
```

Detection: Run with `valgrind ./program` or compile with `-fsanitize=address`:

```
==12345== LEAK SUMMARY:
==12345==    definitely lost: 4 bytes in 1 blocks
==12345==    at 0x4C2FB0F: operator new(unsigned long) (vg_replace_malloc.c:334)
==12345==    by 0x10868B: leaky() (example.cpp:2)
```

The real fix is to never use raw `new/delete`. Use `std::vector`, `std::string`, and smart pointers (Chapter 14), which manage heap memory automatically through RAII.

0.3.23 Stack vs Heap: Which to Use

Use the stack (local variables) when:

- Size is known at compile time
- Lifetime matches the current scope
- Performance is critical (tight loops, small structures)

Examples: `int`, `double`, small structs, `std::array<int, 10>`

Use the heap (via containers or smart pointers) when:

- Size is determined at runtime (user input, file content)
- Lifetime must extend beyond the creating function
- Data is very large (millions of elements)

Examples: `std::vector`, `std::string`, objects stored in maps

In modern C++:

- You almost never write `new/delete` directly
- `std::vector` and `std::string` manage their heap memory internally
- smart pointers (`unique_ptr`, `shared_ptr`) handle ownership automatically
- Raw `new/delete` is a code smell in modern C++

0.3.24 Common Mistakes in This Chapter

0.3.24.1 Mistake 1: `delete` Instead of `delete[]` for Arrays

The bug:

```
int* arr = new int[100];
delete arr;    // undefined behavior! Should be delete[]
```

The symptom: Memory corruption or crash – the allocator uses the wrong size to free. **The fix:** `delete[] arr;`

0.3.24.2 Mistake 2: Double Delete

The bug:

```
int* p = new int{5};
delete p;
delete p;    // undefined behavior -- freeing already-freed memory
```

The symptom: Crash or silent heap corruption. **The fix:** Set pointer to `nullptr` after deleting. Deleting `nullptr` is a no-op.

0.3.24.3 Mistake 3: Using After Delete (Use-After-Free)

The bug:

```
int* p = new int{5};
delete p;
std::cout << *p;    // reads freed memory -- undefined behavior
```

Detection: `-fsanitize=address` reports “heap-use-after-free” with a stack trace.

0.3.25 Exercises

Exercise 8.1 – Where does it live?

For each variable, say whether it lives on the stack or heap:

```
int a = 5;
int* b = new int{10};
std::string s = "hello";
std::vector<int> v = {1, 2, 3};
```

Answer: - a: stack - b: stack (the pointer). The `int` it points to: heap. - s: stack (the `std::string` object itself). The character data it manages: heap (internally). - v: stack (the `std::vector` object). The `int` elements it manages: heap (internally).

Exercise 8.2 – Spot the leak

How many memory leaks does this code have?

```
void process() {
    int* a = new int{1};
    int* b = new int{2};
    if (*a > 0) return;    // early return!
    delete a;
    delete b;
}
```

Answer: Two leaks when `*a > 0` (which is always true here, since `a = 1`). The early return bypasses both `delete` calls. This is exactly why RAII (Chapter 12) and smart pointers (Chapter 14) exist – they clean up even when exceptions or early returns happen.

Exercise 8.3 – Heap array

Allocate an array of 5 doubles on the heap, set them to 1.1, 2.2, 3.3, 4.4, 5.5, print them, then free the memory correctly.

Answer:

```
double* arr = new double[5]{1.1, 2.2, 3.3, 4.4, 5.5};
for (int i = 0; i < 5; ++i)
    std::cout << arr[i] << "\n";
delete[] arr;
arr = nullptr;
```

Chapter 9: const Correctness

0.3.26 Why const Matters

const is not just a safety net for your own mistakes. It is a contract: you tell the compiler “this value will not change,” and the compiler enforces that contract everywhere the value is used. This contract propagates through the codebase, letting the compiler catch bugs at compile time that would otherwise be silent data corruption at runtime.

In Python, there is no const. You use naming conventions (ALL_CAPS) to signal “don’t change this,” but nothing enforces it. C++’s const is enforced.

0.3.27 const Variables

```
const int MAX_PLAYERS = 8;
MAX_PLAYERS = 10;    // COMPILER ERROR: assignment of read-only variable
```

The compiler rejects any assignment to MAX_PLAYERS after initialization. Every future reader of the code knows with certainty: this value never changes.

```
const double PI = 3.14159265358979;
const std::string GREETING = "Hello";
```

Prefer const for anything that does not need to change. It is documentation that is enforced.

0.3.28 const Function Parameters

A const parameter promises not to modify the argument:

```
void print_length(const std::string& s) {
    std::cout << s.size() << "\n";
    s = "modified";    // COMPILE ERROR -- s is const reference
}
```

This is important for const& parameters: you can pass temporaries (rvalues) to them:

```
void print_length(const std::string& s) { ... }

print_length("hello");           // OK: string literal binds to const&
print_length(std::string{"hi"}); // OK: temporary binds to const&

void modify(std::string& s) { ... }
modify("hello");                 // ERROR: non-const reference cannot bind to temporary
```

0.3.29 const Member Functions

When you write a class, member functions that do not modify the object's state should be marked const. This is the const at the end of the function signature:

```
class Rectangle {
public:
    Rectangle(double w, double h) : width{w}, height{h} {}

    double area() const {           // const: does not modify this object
        return width * height;
    }

    void scale(double factor) {     // non-const: modifies this object
        width *= factor;
        height *= factor;
    }

private:
    double width, height;
};

const Rectangle r{3.0, 4.0};
std::cout << r.area();             // OK -- area() is const, usable on const objects
r.scale(2.0);                     // ERROR -- scale() is non-const, not usable on const objects
```

The const after the parameter list means: “this function can be called on const objects, and it promises not to modify the object.”

Rules: - On a const object, only const member functions can be called - A non-const object can call both const and non-const member functions - Inside a const member function, you cannot assign to member variables

```
double area() const {
    width = 0;    // COMPILE ERROR -- modifying member in const function
    return width * height;
}
```

0.3.30 constexpr – Compile-Time Constants

constexpr goes further than const – the value must be computable at compile time:

```
constexpr int BOARD_SIZE = 8;
constexpr double TAX_RATE = 0.085;
constexpr int CELLS = BOARD_SIZE * BOARD_SIZE;    // 64, computed at compile time

// constexpr functions:
constexpr int factorial(int n) {
    return n <= 1 ? 1 : n * factorial(n - 1);
}

constexpr int FACT_10 = factorial(10);    // 3628800, computed at compile time
// The result is embedded in the binary as a constant -- zero runtime cost
```

```
// Cannot be constexpr if value comes from runtime:
int n;
std::cin >> n;
constexpr int x = n;    // ERROR: n is not a compile-time constant
const int y = n;        // OK: const, but runtime value
```

constexpr is useful for: - Mathematical constants (PI, E, conversion factors) - Array sizes (int grid[BOARD_SIZE][BOARD_SIZE]; – array size must be compile-time) - Lookup tables computed at compile time - Performance: zero-cost named constants

0.3.31 Cascading const

const propagates: once you have a const object, everything you do with it must also be const.

```
std::vector<int> v1 = {1, 2, 3};
const std::vector<int> v2 = {4, 5, 6};

v1.push_back(4);    // OK -- v1 is not const
v2.push_back(7);    // ERROR -- v2 is const; push_back is non-const
```

```
int x = v1[0];           // OK
int y = v2[0];           // OK -- operator[] has a const overload (returns const ref)
v2[0] = 99;              // ERROR -- v2[0] returns const int&, cannot assign
```

When you pass by `const&`, you can only call `const` member functions on the object. The compiler checks the entire call chain.

0.3.32 Common Mistakes in This Chapter

0.3.32.1 Mistake 1: Omitting `const` on Member Functions That Should Be `const`

The bug:

```
class Circle {
    double radius;
public:
    double area() { // forgot const
        return 3.14159 * radius * radius;
    }
};

const Circle c{5.0};
c.area(); // ERROR: 'this' argument discards qualifiers (area is non-const)
```

The fix: `double area() const { ... }` – add `const` after the parameter list.

0.3.32.2 Mistake 2: Trying to Modify Through a `const` Reference

The bug:

```
void process(const std::string& s) {
    s += " world"; // ERROR: s is const, cannot modify
}
```

The fix: Take by value (`std::string s`) if you need a modifiable copy, or by non-`const` reference (`std::string& s`) if you intend to modify the original.

0.3.33 Exercises

Exercise 9.1 – Mark const correctly

Which of these should be const members?

```
class Counter {
    int count = 0;
public:
    void increment()    { ++count; }
    int get_count()    { return count; }
    void reset()        { count = 0; }
    bool is_zero()      { return count == 0; }
};
```

Answer: `get_count()` and `is_zero()` should be const – they read but do not modify `count`. `increment()` and `reset()` modify `count`, so they cannot be const.

```
void increment()    { ++count; }
int get_count() const { return count; }
void reset()        { count = 0; }
bool is_zero()      const { return count == 0; }
```

Exercise 9.2 – constexpr table

Write a constexpr function `celsius_to_fahrenheit(double c)` and use it to create a compile-time constant for the boiling point of water (100°C).

Answer:

```
constexpr double celsius_to_fahrenheit(double c) {
    return c * 9.0 / 5.0 + 32.0;
}
constexpr double WATER_BOILING_F = celsius_to_fahrenheit(100.0); // 212.0
```

Chapter 10: Arrays, `std::vector`, and `std::string`

0.3.34 C-Style Arrays (Understand, then Avoid)

The C language had arrays built in. C++ inherited them. They are the underlying model behind everything else, so understand them, then use `std::vector` instead.

```
int scores[5];           // array of 5 ints (uninitialized -- garbage!)
int primes[5] = {2, 3, 5, 7, 11}; // initialized
int zeros[100] = {};     // all zeros (zero-initialized with {})

std::cout << primes[0] << "\n"; // 2 -- zero-indexed
std::cout << primes[4] << "\n"; // 11
```

Memory layout of `primes[5]`:

Address	Value	
0x1000	2	<- <code>primes[0]</code>
0x1004	3	<- <code>primes[1]</code>
0x1008	5	<- <code>primes[2]</code>
0x100C	7	<- <code>primes[3]</code>
0x1010	11	<- <code>primes[4]</code>

The name 'primes' in expressions decays to a pointer to `primes[0]`.

0.3.34.1 Critical Weakness: No Bounds Checking

C++ does not check if your index is valid at runtime:

```
int arr[3] = {1, 2, 3};
std::cout << arr[5]; // reads memory past the array -- undefined behavior
arr[5] = 99;        // writes past the array -- corrupts other data!
```

This is one of the leading causes of security vulnerabilities. The compiler doesn't catch it and there is no runtime exception – it just reads or writes whatever is at that address.

0.3.34.2 Size Is Not Carried With the Array

```
int arr[5] = {1, 2, 3, 4, 5};
sizeof(arr); // 20 -- size in bytes (only works if arr is in scope as an array)

void bad_print(int arr[]) { // array decays to pointer when passed to function
    sizeof(arr);           // 8 -- size of pointer, NOT the array!
}
```

You must pass the size separately, which is error-prone. `std::vector` solves this.

0.3.35 `std::vector` – The Right Way to Do Dynamic Arrays

`std::vector` is C++'s resizable array. It is what Python's `list` most closely corresponds to:

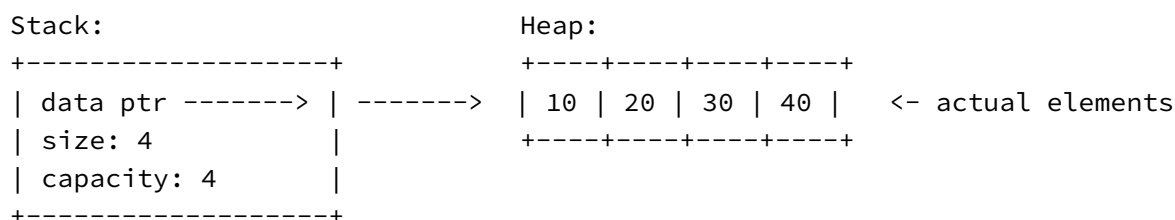
```
# Python list
numbers = [10, 20, 30]
numbers.append(40)
print(len(numbers))    # 4
print(numbers[1])      # 20
```

```
// C++ vector
#include <vector>
std::vector<int> numbers = {10, 20, 30};
numbers.push_back(40);
std::cout << numbers.size() << "\n";    // 4
std::cout << numbers[1] << "\n";       // 20
```

0.3.35.1 What's Inside `std::vector`

A `std::vector` is a small object (24 bytes typically) that internally manages a heap-allocated array:

```
std::vector<int> numbers = {10, 20, 30, 40};
```



- `size`: number of elements currently in the vector
- `capacity`: how many elements fit in the currently allocated heap space
- When `size == capacity` and you `push_back`, the vector allocates a bigger heap array (typically doubles), copies all elements, and frees the old array.

You never see any of this. The vector manages it transparently.

0.3.35.2 Key Operations

```
std::vector<int> v;           // empty vector
v.push_back(10);              // append 10: v = {10}
v.push_back(20);              // append 20: v = {10, 20}
v.push_back(30);              // append 30: v = {10, 20, 30}

v.size();                     // 3    -- number of elements
v.empty();                     // false -- is it empty?
v.front();                     // 10   -- first element
v.back();                      // 30   -- last element
```

```

v.pop_back();           // removes last: v = {10, 20}

v[0];                  // 10 -- no bounds check (unsafe, fast)
v.at(0);               // 10 -- bounds-checked (throws if out of range)

v.insert(v.begin() + 1, 99); // insert 99 at index 1: v = {10, 99, 20}
v.erase(v.begin() + 1);    // remove element at index 1: v = {10, 20}

v.clear();              // remove all elements (size = 0, capacity kept)
v.resize(5, 0);         // resize to 5 elements, new ones initialized to 0

std::vector<int> w(10, 42); // 10 elements all set to 42

```

0.3.35.3 Iterating Over a Vector

```

std::vector<int> v = {1, 2, 3, 4, 5};

// Range-based for (preferred):
for (int n : v)      std::cout << n << " "; // copies each element
for (int& n : v)     n *= 2;                // modifies in place
for (const int& n : v) std::cout << n << " "; // read-only, no copy

// Index-based (when you need the index):
for (int i = 0; i < (int)v.size(); ++i) {
    std::cout << i << ": " << v[i] << "\n";
}

// Note: v.size() returns size_t (unsigned). Comparing int i < size_t is
// technically a warning. Cast to (int)v.size() or use size_t i.

```

0.3.35.4 2D Vectors

```

// 3 rows, 4 columns, all initialized to 0:
std::vector<std::vector<int>> grid(3, std::vector<int>(4, 0));

grid[1][2] = 99;
std::cout << grid[1][2]; // 99

```

0.3.36 std::string – The Right Way to Handle Text

```

# Python strings
s = "hello"
s += " world"
print(len(s))      # 11
print(s.upper())   # HELLO WORLD
print(s[1:5])      # ello

```

```
// C++ strings
#include <string>
std::string s = "hello";
s += " world";
std::cout << s.size() << "\n";          // 11
// (no built-in upper() -- use a loop or std::transform)

std::string sub = s.substr(1, 4);        // "ello" (start=1, length=4)
```

`std::string` internally manages a heap-allocated buffer of characters, similar to `std::vector<char>`.

0.3.36.1 Key Operations

```
std::string s = "Hello, World!";

s.size();                // 13 -- number of characters
s.empty();               // false
s[0];                   // 'H' -- no bounds check
s.at(0);                // 'H' -- bounds-checked
s.front();              // 'H'
s.back();               // '!'

s.find("World");         // 7 -- index where found
s.find("xyz");           // std::string::npos -- not found

s.substr(7, 5);          // "World" (start=7, length=5)

s.replace(7, 5, "C++");  // "Hello, C++!"

std::string t = "hello";
s.compare(t);            // negative (s < t lexicographically)

// Concatenation:
std::string a = "foo", b = "bar";
std::string c = a + b;   // "foobar"
a += "!";               // "foo!"

// Convert to/from numbers:
#include <string>
std::string num_str = std::to_string(42);    // "42"
int n = std::stoi("123");                   // 123
double d = std::stod("3.14");               // 3.14
```

0.3.36.2 String Comparison

Unlike Python where `==` compares value, C++ `std::string` also uses `==` for value comparison:

```
std::string a = "hello", b = "hello";
if (a == b) std::cout << "equal\n";    // equal -- compares content
```

0.3.36.3 Raw C-Style Strings

You will see C-style strings (`const char*`) in older code and C interfaces:

```
const char* cs = "hello";    // points to read-only character array
std::string s = cs;         // convert: C-string to std::string (fine)

// std::string to C-string (for C library functions):
const char* c = s.c_str();  // null-terminated char array
// valid only as long as s is alive and unmodified
```

Always use `std::string` for your own code. Use `const char*` only when a C API requires it, and immediately pass it through – do not store it.

0.3.37 `std::array` – Fixed-Size Array With Safety

When the size is known at compile time, use `std::array` instead of a C-style array:

```
#include <array>
std::array<int, 5> primes = {2, 3, 5, 7, 11};

primes.size();           // 5 -- size is always available
primes[2];               // 5
primes.at(10);           // throws std::out_of_range -- bounds checked
primes.front();          // 2
primes.back();           // 11
```

`std::array` knows its own size, can be passed to functions without separately passing the size, and supports all the standard iteration patterns.

0.3.38 Common Mistakes in This Chapter

0.3.38.1 Mistake 1: Off-By-One / Out-of-Bounds Access

The bug:

```
std::vector<int> v = {1, 2, 3};
for (int i = 0; i <= (int)v.size(); ++i) // <= instead of <
    std::cout << v[i] << "\n";          // v[3] is past the end
```

Detection: –`fsanitize=address` reports “heap-buffer-overflow.” **The fix:** Use `<` not `<=`. Better: use range-based for.

0.3.38.2 Mistake 2: Modifying a Vector While Iterating With Indices

The bug:

```
std::vector<int> v = {1, 2, 3, 4, 5};
for (int i = 0; i < (int)v.size(); ++i) {
    if (v[i] % 2 == 0) v.erase(v.begin() + i);
    // After erasing index 1 (value 2), index 1 now holds 3 (shifted)
    // The loop increments i to 2, skipping 3
}
```

The fix: Iterate backward, or use the erase-remove idiom (Chapter 29).

0.3.38.3 Mistake 3: Using + to Concatenate Non-String Literals

The bug:

```
std::string result = "Count: " + 42; // ERROR: no + between const char* and int
```

The fix:

```
std::string result = "Count: " + std::to_string(42);
// or better (C++23):
std::string result = std::format("Count: {}", 42);
```

0.3.38.4 Mistake 4: Accessing .c_str() Pointer After Modifying the String

The bug:

```
std::string s = "hello";
const char* p = s.c_str();
s += " world"; // may reallocate the internal buffer
printf("%s\n", p); // p may now be dangling
```

The fix: Call .c_str() immediately before passing it to the C function, never store it.

0.3.39 Exercises

Exercise 10.1 – Vector operations

Starting from an empty `std::vector<int>`:

1. Push back 5, 10, 15, 20, 25
2. Print the size (should be 5)
3. Remove the last element
4. Insert the value 99 at index 2
5. Print all elements

Answer:

```

#include <iostream>
#include <vector>

int main() {
    std::vector<int> v;
    v.push_back(5);
    v.push_back(10);
    v.push_back(15);
    v.push_back(20);
    v.push_back(25);

    std::cout << v.size() << "\n";    // 5

    v.pop_back();                      // removes 25; v = {5,10,15,20}

    v.insert(v.begin() + 2, 99);       // v = {5,10,99,15,20}

    for (int n : v)
        std::cout << n << " ";       // 5 10 99 15 20
    std::cout << "\n";
}

```

Exercise 10.2 – String manipulation

Given `std::string sentence = "the quick brown fox"`: 1. Print its length 2. Extract and print the substring “quick” (starts at index 4, length 5) 3. Find where “brown” appears 4. Replace “fox” with “cat” 5. Print the final string

Answer:

```

std::string sentence = "the quick brown fox";
std::cout << sentence.size() << "\n";    // 19
std::cout << sentence.substr(4, 5) << "\n"; // quick
std::cout << sentence.find("brown") << "\n"; // 10

size_t pos = sentence.find("fox");
sentence.replace(pos, 3, "cat");
std::cout << sentence << "\n";           // the quick brown cat

```

Exercise 10.3 – 2D grid

Create a 4x4 grid (vector of vectors), fill position `[row][col]` with the value `row * 4 + col`, then print it as a square.

Answer:

```
std::vector<std::vector<int>> grid(4, std::vector<int>(4));
for (int r = 0; r < 4; ++r)
    for (int c = 0; c < 4; ++c)
        grid[r][c] = r * 4 + c;

for (const auto& row : grid) {
    for (int n : row)
        std::cout << std::setw(3) << n;
    std::cout << "\n";
}
```

Output:

```
0  1  2  3
4  5  6  7
8  9 10 11
12 13 14 15
```

Chapter 11: Scope, Lifetime, and Organizing Code into Files

0.3.40 Scope

Scope is the region of code where a name is visible. In C++, every pair of `{ }` creates a new scope.

```
int x = 1;           // x in outer scope

{
    int x = 2;       // x in inner scope -- SHADOWS the outer x
    std::cout << x;   // 2 -- inner x
}                   // inner x destroyed here

std::cout << x;       // 1 -- outer x, back in scope
```

Shadowing (declaring an inner variable with the same name as an outer one) is legal but confusing. Avoid it. Compile with `-Wshadow` to get warnings.

0.3.40.1 Scope in Loops and Conditionals

```
for (int i = 0; i < 10; ++i) {
    // i only visible inside the loop body
}
// i does NOT exist here

if (int n = get_value(); n > 0) { // C++17 initializer
    // n only visible inside the if/else
}
```

```

} else {
    // n also visible here
}
// n does NOT exist here

```

0.3.41 Lifetime

Lifetime is the period during which a variable's storage is valid. For stack variables, it matches scope. For heap variables, it starts at `new` and ends at `delete`.

```

{
    std::string s = "hello"; // s constructed, storage allocated
    // ... use s ...
} // s destructed, storage freed
// using s here is undefined behavior (it no longer exists)

```

The key insight: in C++, when a variable goes out of scope, its **destructor** is called automatically. For `std::string`, the destructor frees the heap-allocated character buffer. For `std::vector`, it frees the element array. This automatic cleanup is called **RAII** (Chapter 12) and is C++'s answer to garbage collection.

0.3.42 Organizing Code: The Header/Source Split

As programs grow, you split code into multiple files. The organization:

```

projectname/
  include/
    math_utils.h      <- declarations (the public interface)
  src/
    math_utils.cpp    <- definitions (the implementation)
    main.cpp          <- uses the functions

```

0.3.42.1 Why Headers Exist

C++ compiles each `.cpp` file separately. When `main.cpp` calls `multiply(3, 4)`, the compiler must know (at compile time) what `multiply` looks like – its parameter types and return type – to type-check the call.

The linker resolves where `multiply`'s actual code is. But the compiler must see the declaration.

The solution: put declarations in a header file. Every `.cpp` that needs to call `multiply` includes the header.

```
// math_utils.h -- declarations only
#pragma once

int add(int a, int b);
int subtract(int a, int b);
int multiply(int a, int b);
double divide(double a, double b);
```

`#pragma once` tells the preprocessor: include this file at most once per compilation unit, even if it is `#include`d multiple times. It prevents duplicate declaration errors from diamond-shaped include chains.

```
// math_utils.cpp -- definitions
#include "math_utils.h" // include own header to verify declarations match

int add(int a, int b)      { return a + b; }
int subtract(int a, int b) { return a - b; }
int multiply(int a, int b) { return a * b; }
double divide(double a, double b) { return a / b; }
```

```
// main.cpp -- uses the functions
#include <iostream>
#include "math_utils.h" // gets the declarations; tells compiler the signatures

int main() {
    std::cout << add(3, 4) << "\n"; // 7
    std::cout << multiply(6, 7) << "\n"; // 42
    std::cout << divide(10.0, 4.0) << "\n"; // 2.5
}
```

Build:

```
$ g++ -std=c++23 -Wall -o program src/main.cpp src/math_utils.cpp -I include/
```

The `-I include/` tells the compiler to look in the `include/` directory for headers.

0.3.42.2 The Compilation Model With Multiple Files

```
math_utils.cpp --> [compiler] --> math_utils.o  --+
main.cpp       --> [compiler] --> main.o        --+--> [linker] --> program
```

Each `.cpp` is compiled independently. The compiler only needs to see the header (declarations) to compile `main.cpp`. It trusts that `math_utils.cpp` will provide the actual definitions. The linker connects everything.

0.3.42.3 What Goes In Headers vs Source Files

Headers (.h):

- Function declarations
- Class declarations (but not the bodies of non-inline methods)
- Type definitions (structs, enums, using aliases)
- #include directives needed by the declarations
- inline functions (small utility functions that are called often)
- constexpr and const variable definitions

Source (.cpp):

- Function definitions (the actual code)
- Class method definitions
- Global variable definitions
- #include of their own header + other headers needed for the implementation
- Implementation details not exposed to users

Do NOT put in headers: - using namespace std; - it pollutes every file that includes your header - Non-inline function definitions - causes “multiple definition” linker errors if included in multiple .cpp files - Definitions of non-const global variables - also causes linker errors

0.3.42.4 Include Guards (Alternative to #pragma once)

Older code uses include guards instead of #pragma once:

```
#ifndef MATH_UTILS_H
#define MATH_UTILS_H

// ... declarations ...

#endif // MATH_UTILS_H
```

If math_utils.h is included twice in the same compilation unit, the second inclusion sees that MATH_UTILS_H is already defined and skips everything inside. #pragma once does the same thing more concisely and is supported by all modern compilers.

0.3.43 Namespaces

Namespaces prevent name collisions between libraries. When two libraries both define Matrix, they can put it in different namespaces.

```

namespace geometry {
    struct Point { double x, y; };
    double distance(Point a, Point b);
}

namespace graphics {
    struct Point { float x, y, z; }; // different Point, no conflict
    void draw(Point p);
}

// Use with ::
geometry::Point p1{1.0, 2.0};
graphics::Point p2{3.0f, 4.0f, 5.0f};

```

The standard library lives in `std`. That is why everything is `std::cout`, `std::vector`, `std::string`.

You can pull specific names into scope:

```

using std::cout; // just cout, not everything
using std::vector;

cout << "hello\n"; // OK
vector<int> v; // OK

```

Do this inside functions, not at file scope (it affects the rest of the file, which may cause surprises in headers).

0.3.43.1 Writing Your Own Namespace

```

// mylib.h
#pragma once

namespace mylib {

int clamp(int value, int lo, int hi);
double lerp(double a, double b, double t);

} // namespace mylib

// mylib.cpp
#include "mylib.h"

namespace mylib {

int clamp(int value, int lo, int hi) {
    if (value < lo) return lo;
    if (value > hi) return hi;
    return value;
}

double lerp(double a, double b, double t) {
    return a + t * (b - a);
}

```

```
}  
  
} // namespace mylib
```

0.3.44 Static Local Variables

A static local variable is initialized once and persists across calls:

```
int counter() {  
    static int count = 0; // initialized only once (first call)  
    ++count;  
    return count;  
}  
  
std::cout << counter() << "\n"; // 1  
std::cout << counter() << "\n"; // 2  
std::cout << counter() << "\n"; // 3
```

The variable `count` lives for the entire program duration (like a global), but is only accessible inside `counter`. This is useful for functions that need to remember state between calls without using a class.

0.3.45 Common Mistakes in This Chapter

0.3.45.1 Mistake 1: Defining a Non-Inline Function in a Header

The bug:

```
// utils.h  
int add(int a, int b) { return a + b; } // DEFINITION in header  
  
// main.cpp includes utils.h  
// utils.cpp also includes utils.h  
// Both .cpp files define add() -- linker error!
```

linker error: multiple definition of `'add(int, int)'`

The fix: Declarations in headers, definitions in `.cpp` files. Or mark the function `inline` (which tells the linker to allow multiple identical definitions).

0.3.45.2 Mistake 2: Missing `#pragma once` (or Include Guard)

The bug:

```
// a.h includes b.h and c.h
// b.h includes c.h
// c.h has no include guard

// When a.h is processed: c.h is included twice, all declarations in c.h appear twice
error: redefinition of 'class Foo'
```

The fix: Start every header with `#pragma once`.

0.3.45.3 Mistake 3: using namespace std; in a Header

The bug:

```
// utils.h
#include <string>
using namespace std;    // pollutes every file that includes utils.h

// Now every file including utils.h gets all of std:: imported
// Causes conflicts with user-defined names like 'vector', 'string', 'max', 'min'
```

The fix: Never put `using namespace X;` in a header. In `.cpp` files it is acceptable (though `using std::cout;` for specific names is better style).

0.3.46 Exercises

Exercise 11.1 – Trace scope

Predict the output:

```
int x = 10;
{
    int x = 20;
    {
        int x = 30;
        std::cout << x << "\n";
    }
    std::cout << x << "\n";
}
std::cout << x << "\n";
```

Answer: 30, 20, 10. Each inner `x` shadows the outer one. When the inner scope ends, the outer `x` becomes visible again.

Exercise 11.2 – Split into files

Split this single-file program into `main.cpp`, `greeting.h`, and `greeting.cpp`:

```
#include <iostream>
#include <string>

std::string make_greeting(const std::string& name) {
    return "Hello, " + name + "!";
}

int main() {
    std::cout << make_greeting("Alice") << "\n";
    std::cout << make_greeting("Bob") << "\n";
}
```

Answer:

```
// greeting.h
#pragma once
#include <string>

std::string make_greeting(const std::string& name);
```

```
// greeting.cpp
#include "greeting.h"

std::string make_greeting(const std::string& name) {
    return "Hello, " + name + "!";
}
```

```
// main.cpp
#include <iostream>
#include "greeting.h"

int main() {
    std::cout << make_greeting("Alice") << "\n";
    std::cout << make_greeting("Bob") << "\n";
}
```

Build: `g++ -std=c++23 -o program main.cpp greeting.cpp`

Exercise 11.3 – Static local counter

Write a function `next_id()` that returns 1 on the first call, 2 on the second, and so on, without using any global variable.

Answer:

```
int next_id() {
    static int id = 0;
    return ++id;
}
// next_id() == 1, next_id() == 2, next_id() == 3 ...
```

Exercise 11.4 – Namespace

Write a namespace `convert` with two functions: `km_to_miles(double km)` and `miles_to_km(double miles)`. 1 km = 0.621371 miles.

Answer:

```
// convert.h
#pragma once

namespace convert {
    double km_to_miles(double km);
    double miles_to_km(double miles);
}
```

```
// convert.cpp
#include "convert.h"

namespace convert {
    double km_to_miles(double km)    { return km * 0.621371; }
    double miles_to_km(double miles) { return miles / 0.621371; }
}
```

Usage:

```
std::cout << convert::km_to_miles(100.0) << "\n";    // 62.1371
std::cout << convert::miles_to_km(62.0) << "\n";    // 99.79...
```

Part II is complete. You now understand the concepts that have no Python equivalent: references as aliases, pointers and memory addresses, the stack vs heap memory model, const correctness enforced by the compiler, the standard library containers, and how C++ organizes multi-file projects.

Part III covers ownership and memory management – the RAII pattern, smart pointers, and move semantics. These are what make modern C++ safe while staying fast. Ask to continue.